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Mean based relief: An improved feature selection method based on ReliefF

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Abstract
Selection of relevant features is vitally important in machine learning tasks involving large datasets with numerous features. It helps in reducing the dimensionality of a dataset and improving model performance. This study introduces a feature selection technique named μ-Relief, which is based on ReliefF, one of the most extensively used Relief-based algorithms. μ-Relief effectively determines the most relevant feature subset and significantly outperforms the ReliefF algorithm. ReliefF estimates feature quality considering only the nearest neighbors, resulting in low classification accuracy on non-uniformly distributed or noisy datasets. The proposed μ-Relief technique considers neighbors with more effective information on the basis of mean distance. It utilizes neighbors far from the mean distance to obtain feature weight estimates, which improves the algorithm's performance. The algorithm was tested on thirteen real-world datasets and validated on three synthetic datasets. Its effectiveness in selecting relevant features was evaluated by comparing it to other well-known feature selection algorithms, namely Chi-Square, ANOVA, MI, CMIM, MRMR, SURF*, MultiSURF, MultiSURF*, and ReliefF. When evaluated using multiple classifiers trained on the features selected by different feature selection techniques, the metrics of classification accuracy, weighted F1-score, and ROC-AUC, showed that μ-Relief effectively determined relevant features and outperformed other techniques.

Keywords Feature selection · Filter method · Machine learning · ReliefF

1 Introduction

The ubiquity of AI and IoT has resulted in the rapid generation and aggregation of an enormous amount of data. Although big data is of great value and essential for the advancements of AI, its high dimensionality leads to several new challenges, such as more storage and computational capacity [1]. Big data cannot be organized, stratified, and scrutinized in lesser dimensions. Moreover, the algorithms developed for lower dimensional space work inefficiently in high dimensional space as a large number of features tend to overfit classification models. In a dataset with a large number of features, not all features have relevant information, known as the curse of dimensionality [2]. Therefore, it becomes crucial to filter irrelevant and redundant features by selecting a suitable subset of relevant features that helps avoid overfitting, enhance predictive accuracy and make models computationally inexpensive. Frequently used dimension reduction techniques are broadly categorized as i) Feature Selection and ii) Feature Extraction. A subset of original features is selected based on some statistical test or estimate by an ML algorithm in the feature selection process. The feature's physical meaning or its existence remains unaltered, giving models better readability and interpretability [3, 4]. In the feature extraction technique, a new feature space is generated using two or more features. As the new feature space does not retain the physical meaning of the original feature set, further investigations become difficult while using this technique [5]. However, both methods lessen the dimensions, improve the performance of ML models, prevent overfitting, and minimize the computational cost and storage space.

Feature selection techniques are divided into three categories: i) filter methods [6], ii) wrapper methods [7] and iii) embedded methods [8]. In the filter approach, a feature subset is selected based on the properties of the data itself. Most filter methods use different statistical techniques to calculate

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a score for each feature and shortlist the features with the highest scores. However, a few algorithms select features iteratively in a greedy forward fashion. The feature subset is chosen without involving any ML algorithm making filter methods computationally less expensive. This makes filter methods more suitable for high-dimensional data analysis; however, in the absence of specific learning guidance, the chosen feature subset, at times, is not optimal. The wrapper feature selection approach creates different subsets of features and selects those that yield the best-performing model based on specific performance metrics. Since the model has to learn the feature space of each subset of features, it is prone to overfitting, and as a result, the wrapper selection process is computationally intensive. Hence, wrapper methods are more suited for low-dimensional data. Other than supervised ML models, wrapper techniques also use unsupervised methods such as clustering [9], and binary variants of metaheuristic algorithms [10] such as evolution-based [11] and swarm intelligence-based algorithms [12]. Evolution-based algorithms such as the genetic algorithm [13] initiate their processes with a randomly generated solution space and result in binary vector representations that are used to make decisions to select or deselect each feature. Swarm intelligence-based algorithms such as whale optimization [14], dragonfly algorithm [15], grey wolf optimization [16], black hole algorithm [17, 18] and prairie dog optimization [19] are each based on diverse swarm behaviors. When used for feature selection, the binary forms of these algorithms compute binary solution vectors, where selection of a particular feature is indicated by 1, and 0 implies that the corresponding feature is not considered in the feature subset. However, the performances of these algorithms heavily depend on the underlying fitness function and other algorithm-specific parameters, thereby incurring higher computational costs as compared to filter methods. Embedded methods are hybrid models that have the characteristics of both filter and wrapper methods, and integrate feature selection with model learning. In these algorithms, optimal feature subsets are obtained based on data relevance, and then cross-validation is performed to decide the best subset. They incorporate interactions with the learning algorithms and do not evaluate feature subsets iteratively like wrapper methods. Therefore, their time complexities are closer to those of filter methods.

The Relief algorithm is a filter-type feature selection method that considers the nearest neighbor for the estimation of weights and applies only to binary classification problems with discrete features [20, 21]. In this algorithm, weights are not optimized efficiently as only one nearest neighbor is considered for feature quality estimation. However, Relief derivatives have also been proposed in the literature that work with continuous features, deal with missing and imbalanced data, and can be applied to regression and multi-class classification. For instance, ReliefF deals with missing data efficiently and is applicable to multi-class problems [22]. Since it considers only the nearest neighbors for weight estimation, only sub-optimum weights can be estimated because of noise and irrelevant features. This algorithm is considered shortsighted [23], as incorporating more neighbors makes the weight estimations random and reduces its ability to select relevant features. ReliefF variant named RReliefF [24], formulated for regression problems, uses relative distances between the predicted values of pairs of instances and is computationally more expensive than ReliefF. Spatially Uniform ReliefF (SURF) defines nearest neighbors based on threshold distance and detects two-way epistatic interactions. This algorithm uses more neighbors for weight estimation in densely populated spaces. However, in sparse regions, only a few neighbors are used, leading to suboptimal weight estimation. This technique is limited to continuous features, and cannot examine gene and environment interactions [25]. SURF* considers both informative extremes, i.e., the nearest and farthest neighbors for each sample It uses a step function which generates large changes in quality estimates of attributes, often misleading the optimization of weights [26]. The SWRF* [27] extends the SURF* algorithm by using sigmoid weighting to avoid the large changes in weight estimation and detect two-way epistatic interactions. MultiSURF* was designed to improve the time efficiency of the SURF* algorithm [28]. However, MultiSURF* inherits the shortcomings of SURF* because the underlying methodology used to measure feature quality in both methods is the same. The study in [29] claims that there is no evidential support for far-scoring on 3-way epistatic interactions. It introduces MultiSURF, which disregards the far-scoring mechanism but preserves the dead-band zone and mechanism of target instance-centric neighbor identification from MultiSURF*. The number of hits and misses for each target class may be imbalanced in this approach because neighbours are selected based on a threshold. The core Relief algorithm was reformulated by incorporating variance of the nearest neighbor distances for quality estimation of attributes in the STIR [30] algorithm. Though it works only for binary classification, the consideration of variance in this algorithm allows the adjustment for multiple testing of Relief-based scores and statistical significance calculation of features. LLH-Relief [31] uses L1-norm regularization and logistic regression to solve an optimization problem to estimate feature weights. Since this method uses logistic regression, it inherits the time complexity issues associated with ML-based feature selection algorithms. The ED-Relief [32] algorithm uses a novel effective distance measure to handle the co-occurrence of within and between class imbalance. The performance of this technique is good for a small feature set, but it degrades with more features and is inconsistent when the dataset contains numeric features. MRelief [33] is a multidirectional neighbor

search technique to find all neighbors within a distance limit with the consideration of different orientations. As a result, the weights estimated are more optimal than those provided by Relief. However, it is highly computationally expensive when used with high-dimensional and large datasets. This study introduces μ-Relief, a filter method of feature selection, that uses mean distances to gather relevant information from the farthest and the nearest k neighbors to estimate feature relevance. This method uses the two extremes of the neighbourhood of each randomly selected sample to estimate weights, which captures the diversity of information inherent in each feature. The proposed algorithm yields optimal feature subsets for non-uniformly distributed feature values as well. This algorithm is sensitive to complicated patterns of interactions amongst features and flexible in consistently adapting to different data characteristics requirements. It is computationally efficient for both high-dimensional feature space and large-scale datasets. In the present study, the proposed algorithm has been implemented and tested on different datasets with multiple types of features, namely, categorical and numerical, having issues such as class imbalance, for binary and multi-class classification problems.

The remainder of this paper is organized as follows: Section 2 describes filter-based feature selection methods. The proposed algorithm is presented with its asymptotic analysis in Section 3. Experimental results on synthetic and real-world datasets, relevant discussion of the performance, strengths, and weaknesses of the algorithm have been discussed in Section 4. Finally, Section 5 offers conclusions, usage recommendations and direction for future work.

2 Theoretical background

2.1 Filter methods

As previously described, in the filter method, a statistical score is calculated to rank features or feature subsets for quality estimation. Filter methods are further classified into two categories: univariate and multivariate. In univariate filter methods such as Analysis Of Variance (ANOVA), Variance and Chi-square tests, each feature is evaluated individually and feature interactions are not considered. In contrast, in multivariate filter methods like Mutual Information (MI) and Maximum Relevance - Minimum Redundancy (MRMR), features are evaluated in the context of other features.

2.2 Relief-based algorithms

Relief and its derivatives are individual evaluation filter methods. Instead of inferring feature interactions using statistical measures of feature combinations, most RBAs use nearest neighbor instances to calculate weight estimates that account for feature interactions indirectly [34]. Hence, these algorithms have the ability to detect interactions in two dimensions only. However, few RBAs can detect higher-order interactions, such as 3-way or 4-way. In addition to being used for the selection of best features, RBAs can also provide specialized information to train stochastic ML algorithms such as evolutionary algorithms.

2.3 Relief

The original Relief algorithm, formulated by Kira and Rendell [20, 21], was inspired by instance-based learning [35] and was limited to binary classification problems. Relief is non-myopic [36], as it measures the quality of a given feature considering other features. It is also non-parametric [37], as sample size and population distribution are not considered. In Relief, the quality of an attribute is estimated on the basis of the difference between the feature values of a randomly selected query point of one class and its nearest neighbors from the same and different classes. If the same class neighbor has a non-zero difference with the query point, a penalty is applied to the weight estimate of the feature in consideration. If the neighbor belongs to the other class, a reward is applied to the quality estimate of the same feature. However, selected neighbors and estimated weights in Relief tend to become unstable as the feature set size increases [38].

2.4 ReliefF

An improved Relief derivative known as ReliefF is the most widely used variant [39] in practical applications. It can handle multi-class classification problems and works well for noisy datasets with missing values. It uses an interpolation approach to handle missing values. The class-conditional probability of two instances having different values for the feature under consideration is used as the value of difference when a missing value is encountered. It considers the k nearest neighbors from each class to estimate the feature weights. With this improvement, the reliability of weight estimation increases, especially for noisy datasets. For weight estimation, contributions of the same class (hits), and all the other classes (misses), are considered using the respective class-wise probabilities. Ideally, this motivates the algorithm to estimate the quality of features to separate each class regardless of inter-class distances [22]. It inherited the qualities of the variants proposed in ReliefA to ReliefE, and was considered the better approach.

3 Proposed algorithm

The new RBA proposed in this study is called μ-Relief. It preserves all aspects of ReliefF except the process of identifying

neighbors for determining feature scores. Unlike ReliefF, it uses mean distances to extract effective information from the farthest and the closest k neighbors. To understand graphically, consider Fig. 1, showing the data points plotted on a two-dimensional feature space. The data point O has values (0, 0) and other data points A, B, C, D, E and F have different feature values. Though Manhattan distance-wise, all the points are equidistant from O, but feature-wise, they carry different information, as depicted by their spatial arrangement in the 2-D space.

Consider a feature vector x_i and its two same-class neighbors, x_j and x_p in a high-dimensional space comprising N dimensions.

$$x_i = < a_1 \quad a_2 \quad a_3 ..... a_{N-1} \quad a_N > \tag{1}$$

$$x_j = < a_1 \quad a_2 \quad a_3 ..... a_{N-1} \quad a_N + \epsilon > \tag{2}$$

$$x_p = < a_1 + \epsilon \quad a_2 + \epsilon \quad a_3 + \epsilon ..... a_{N-1} + \epsilon \quad a_N + \epsilon > \tag{3}$$

The nearest and farthest neighbors of x_i are assumed to be x_j and x_p, respectively. As shown in Eqs. (1) and (2), x_i and x_j have the same feature vectors till the $(N-1)^{th}$ dimension, only differing by ϵ in their last dimensions, with $\epsilon << a_k$ for all k ranging from *1 to N*, limits inclusive. The data point x_j reinforces redundant information in $(N-1)$ dimensions. As a result, the approach of using the closest neighbor only for estimating feature relevance does not take into account the variation in the feature values of the instances of the same class. On the other hand, as shown in Eqs. (1) and (3), the farthest neighbor x_p differs from x_i by ϵ in each dimension. Hence, the cumulative distance between them is comparatively larger than the distance between x_i and x_j. Therefore, x_i and x_p do not reinforce the repetitive information. However, considering x_p as a neighbor makes the neighbor selection approach inclusive of the variation in the feature values of the same class. The feature weights, thus computed, on the basis of the nearest and farthest neighbors capture the diversity of information in each class.

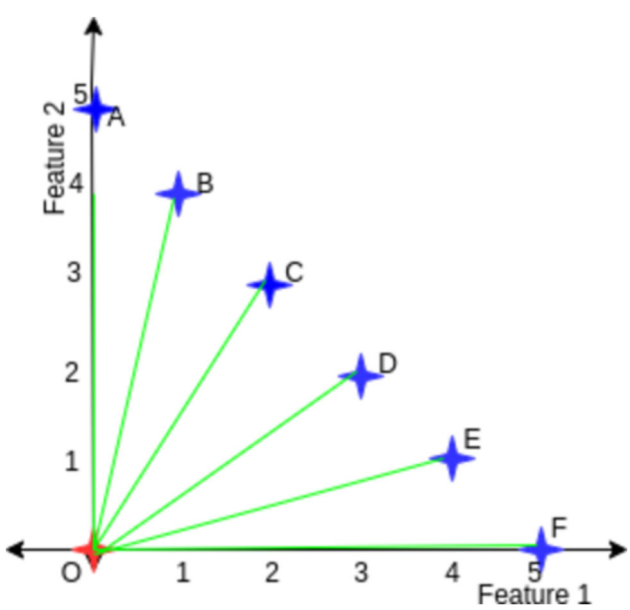

Fig. 1 A schematic example of instances in two dimensional feature space

3.1 Algorithm description

Consider a dataset $X = \{x_1, x_2, ..., x_m\}$ having m data instances in R^N, where each instance has N features. The ith instance can be defined as $x_i = < F_{1i} \; F_{2i} \; ..... \; F_{Ni} >$ where F_{pi} represents the pth feature value of the instance. The corresponding class vector for dataset X is defined as $Y = < y_1 \; y_2 ..... \; y_m >$. As in the ReliefF algorithm, μ-Relief selects an instance x_i randomly where $x_i \in X$. Hit and Miss sets are obtained according to y_i. In the algorithm, the set of instances of the same target class as x_i is known as Hit (H), where $H \subseteq X$. A set of instances that belong to any class except y_i, is known as Miss denoted by M_j where $j \in Y$ and $j \neq y_i$. Further, distance vectors are calculated using Manhattan distances between x_i and all instances of H, as well as x_i and all instances of M_j separately. The Manhattan distance between x_i and any instance x_r, is calculated using the following equation.

$$md(x_i, x_r) = \sum_{q=1}^{N} \| F_{qi} - F_{qr} \| \tag{4}$$

Here, F_{qi} and F_{qr} are the qth feature values of x_i and x_r, respectively. The distance vector between x_i and H is evaluated using the following equation.

$$D_{iH} = < d_{i1} \quad d_{i2} \quad d_{i3} \quad ... d_{ia} \quad ... d_{ih} > \tag{5}$$

Here, h is the number of instances in H, a is an integer between 1 and h (limits inclusive). If H_a is an instance in H, then, d_{ia} is defined by Eq. (6).

$$d_{ia} = md(x_i, H_a) \tag{6}$$

Similarly, the distance vector between x_i and M_j is evaluated using the following equation.

$$D_{iM_j} = < d_{i1} \quad d_{i2} \quad d_{i3} \quad ... d_{ia} \quad ... d_{im_j} > \tag{7}$$

Here, m_j is the number of instances in M_j, a is an integer between 1 and m_j (limits inclusive). If M_{ja} is an instance in M_j, then, d_{ia} is defined by Eq. (6).

$$d_{ia} = md(x_i, M_{ja}) \tag{8}$$

Novel mean-based approach for neighborhood selection The selection of neighborhoods is a decisive step in RBAs. The proposed method uses a different approach to select k neighbors from each class. The proposed process is described as follows:

- First, the elements of each distance vector are replaced by the element-wise absolute difference with the mean of the distance vector.
- Second, elements of each distance vector are sorted in the non-increasing order.
- Third, the top k neighbors from each distance vector are selected for further consideration in the feature selection process.

In the ReliefF algorithm, only nearest neighbors are selected that may be less relevant, redundant or noisy [26]. However, in the proposed approach, using both extremes in the neighborhood results in more relevant features that capture the information diversity in each attribute. Since the selected neighbors are not required to be the distance-wise closest neighbors, redundant information induced by nearby neighbors is also reduced. This modified process of selecting neighborhoods for a random instance emphasizes more on the instances far from the mean of the distances. It also regards the instances closer to the mean of the distances as the least informative. The distance distribution between a random query point and selected neighborhood must have enough contrast [40]. Consider μ_{iH} as the mean of D_{iH}. Then, the new distance vector D^*_{iH} is given by the following equation.

$$D^*_{iH} = <(d_{i1}-\mu_{iH})(d_{i2}-\mu_{iH})(d_{i3}-\mu_{iH}) \\ ...(d_{ia}-\mu_{iH})...(d_{ih}-\mu_{iH})> \tag{9}$$

The same calculations are performed with respect to each Miss set M_j to obtain $D^*_{iM_j}$. The next step towards feature quality estimation is to calculate the values of difference (diff) between x_i and each of its k neighborhood instances per class for each feature F_p. This calculation for numerical features is governed by Eq. (10).

$$diff(F_p, x_i, x_r) = \frac{|F_{pi} - F_{pr}|}{[max(F_p) - min(F_p)]} \tag{10}$$

The calculation of differences for categorical features is governed by Eq. (11).

$$diff(F_p, x_i, x_r) = \begin{cases} 0: & F_{pi} = F_{pr} \\ 1: & otherwise \end{cases} \tag{11}$$

The differences calculated using Eqs. (10) and (11) are used to compute feature weights (quality estimates). Weights of all features are initialized to zero. The values of difference with the hit instances are added to the corresponding feature weights since a positive difference with the same class instances for a feature implies that this feature is unfavorably discriminating same class instances. On the contrary, values of difference with the miss instances are subtracted from the corresponding feature weights. A positive difference with the instances of other classes for a feature implies that this feature is favorably discriminating instances of different classes.

Algorithm 1 μ-Relief.

Require: $N \geq 1 \vee m \neq 0$
Ensure: $X \in R^N$
$W[N] \Leftarrow 0$
$i \Leftarrow 1$
while $i \leq m$ **do**
  $x_i = random(X)$
  $find(H)$
  $find(M_j)$
  $D_H, D_{M_j} = md(x_i, x_r)$
  $compute(\mu_{iH}), compute(\mu_{iM})$
  $D^*_H = D - \mu_{iH}, D^*_M = D - \mu_{iM}$
  $farthest_neighbour(D^*_H, k), farthest_neighbour(D^*_M, k)$
  a = 1
  while $a \leq N$ **do**
$$W[N] = W[N] - \sum_{l=1}^{k} diff(a, x_i, H_l)(m * k) + \sum_{c \neq class(x_i)} \frac{P(c)}{1 - P(class(x_i))} \sum_{l=1}^{k} diff(a, x_i, M_{jl})(m * k)$$
    $a \Leftarrow a + 1$
  end while
  $i \Leftarrow i + 1$
end while

Computational complexity The μ-Relief method iterates for m iteration where m is a user-defined value, ranging at max total number of instances. As it has the upper limit of M (total number of instances in the dataset), one loop for N features and one loop for k neighbors, its computational time is $O(mkN)$. The selection of neighbors also required distance calculation between x_i (random instance) and all other instances of each class. Compared to other ReliefF methods, the time complexity of the proposed method is comparable.

4 Results and analysis

This section presents the evaluation results of the μ-Relief on three synthetic datasets and thirteen real-world datasets. *Python* was the programming language of choice for the experiments. Implementations of Chi-Square, ANOVA, Mutual Information, MRMR and CMIM (Conditional Mutual Information Maximization criterion) in *scikit-learn's* feature selection module and *scikit-feature* package for the

analysis of non-Relief-based algorithms were used. The ReliefF, SURF, SURF*, MultiSURF and MultiSURF* algorithms were implemented using the *scikit-rebate* package for RBA experiments. All the classification algorithms used were selected from the *scikit-learn* library, except XGBRF (XGBoost with Random Forest), which was selected from the *xgboost* library.

4.1 Experiments on synthetic datasets

Using synthetic datasets, an algorithm can be tested in controlled settings. Furthermore, the standard behavior of an algorithm can be studied under regulation and its validity as a standard feature selection algorithm can be confirmed. In this study, three synthetic datasets were generated to understand the specific behaviors of the proposed algorithm and evaluate its ability to assign better quality estimates to features as compared to other algorithms. The corresponding three experiments are described in the following sections.

4.1.1 Experiment 1

This experiment was conducted to verify the behavior of μ-Relief to select informative features and reject features of less or no relevance. As shown in Table 1, a synthetic dataset $E_1 = (I, F, T)$ was designed for the experiment, where, I represents an instance, $F = \{F_1, F_2, F_3\}$ is the feature set, and T is the target class. T is dependent on F_1 only and F_2, F_3 are inconsequential features to predict the target. Since the target is binary, the set of hits and misses are denoted by H and M, respectively. The following assumptions were made to run the μ-Relief algorithm on E_1. Two iterations (m = 2) were used to compute the weights, I_4 and I_6 were the randomly selected instances in iterations 1 and 2, respectively, and two neighbors (k = 2) were considered for weight estimation. Since, I_4 belongs to target class 1, the hit set, $H = \{I_1, I_2, I_3\}$, and miss set, $M = \{I_5, I_6, I_7, I_8\}$.

Table 1 Synthetic Dataset E_1

Instance(I)	F_1	F_2	F_3	T
I_1	1	1	1	1
I_2	1	1	0	1
I_3	1	0	1	1
I_4	1	0	0	1
I_5	0	1	1	0
I_6	0	1	0	0
I_7	0	0	1	0
I_8	0	0	0	0

The following distances vectors were calculated using Eqs. (4), (5), (6) and (7).

$$\begin{aligned} D_{4H} &= \quad < d_{41} \quad d_{42} \quad d_{43} > \\ &= \quad < 2 \quad 1 \quad 1 > \end{aligned} \tag{12}$$

$$\begin{aligned} D_{4M} &= \quad < d_{45} \quad d_{46} \quad d_{47} \quad d_{48} > \\ &= \quad < 3 \quad 2 \quad 2 \quad 1 > \end{aligned} \tag{13}$$

The values of μ_{4H} and μ_{4M} are 1 and 2, respectively. Therefore, the new distance vectors, D^*_{4H} and D^*_{4M} (Eqs. (14) and (16)), were obtained using Eq. (9).

$$D^*_{4H} = \quad < 1 \quad 0 \quad 0 > \tag{14}$$

$$D^*_{4M} = \quad < 1 \quad 0 \quad 0 \quad 1 > \tag{15}$$

Next, D^*_{4H} and D^*_{4M} were sorted in non-increasing order as described in Algorithm 1. Since $k = 2$, $\{I_1, I_2\}$ and $\{I_5, I_8\}$ were selected from H and M, respectively for weight (quality) estimations. The computed weights at the end of the first iteration, W^1 are shown in Eq. (17).

$$W^1 = [W^1_{F_1} \quad W^1_{F_2} \quad W^1_{F_3}] \tag{16}$$

Here, $W^1_{F_1} = 0.50$, $W^1_{F_2} = -0.25$, and $W^1_{F_3} = 0.00$. Therefore, it can be observed that only F_1 has a positive value of weight after the first iteration. Similarly, weight estimations are updated in the second iteration using the computations shown in Eqs. (18) through (22). The values of μ_{6H} and μ_{6M} are 1 and 2, respectively.

$$\begin{aligned} D_{6H} &= \quad < d_{65} \quad d_{67} \quad d_{68} > \\ &= \quad < 1 \quad 2 \quad 1 > \end{aligned} \tag{17}$$

$$\begin{aligned} D_{6M} &= \quad < d_{61} \quad d_{62} \quad d_{63} \quad d_{64} > \\ &= \quad < 2 \quad 1 \quad 3 \quad 2 > \end{aligned} \tag{18}$$

$$D^*_{6H} = \quad < 0 \quad 1 \quad 0 > \tag{19}$$

$$D^*_{6M} = \quad < 0 \quad 1 \quad 1 \quad 0 > \tag{20}$$

$$W^2 = [W^2_{F_1} \quad W^2_{F_2} \quad W^2_{F_3}] \tag{21}$$

Here, $W^2_{F_1} = 1.00$, $W^2_{F_2} = -0.25$, and $W^1_{F_3} = -0.25$. It can be observed that only F_1 has a positive value of weight after the second iteration. Hence it can be concluded that F_1 is the only informative feature for target class T, whereas F_2 and F_3 are features of less or no importance.

4.1.2 Experiment 2

This experiment was designed to validate the far-sightedness of the μ-Relief algorithm and compare its ability to select relevant features with other filter methods of feature selection.

Table 2 Features selected using each method in Experiment-2

Method	Top-3 Features	Number of features selected
Chi-Square	$\{A_2, A_3, A_4\}$	2
ANOVA	$\{A_1, A_3, A_2\}$	3
MI	$\{A_3, A_2, A_4\}$	2
MRMR	$\{A_9, A_{10}, A_8\}$	0
CMIM	$\{A_4, A_5, A_3\}$	1
ReliefF	$\{A_2, A_3, A_4\}$	2
SURF*	$\{A_1, A_2, A_3\}$	3
MultiSURF	$\{A_2, A_3, A_1\}$	3
MultiSURF*	$\{A_1, A_2, A_3\}$	3
μ-Relief	$\{A_2, A_3, A_1\}$	3

A toy dataset $E_2 = (I, F, T)$ was synthetically generated for the experiment using a parity-like problem. Here, $F = \{A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8, A_9, A_{10}\}$, and target binary class T is defined by Eq. (23).

$$T = ((A_1 \oplus A_2) \wedge (A_1 \wedge A_3)) \tag{22}$$

Equal number of instances of both classes (0 and 1) were randomly selected to obtain a dataset of size 200 from $1024(2^{10})$ possible combinations of feature values. Chi-Square, ANOVA, MI, MRMR, CMIM, ReliefF, SURF*, MultiSURF, MultiSURF*, and μ-Relief methods were used for feature selection. The target variable was computed using three features only, namely, A_1, A_2 and A_3. The three most relevant features concluded by each method have been reported in Table 2. The results in Table 2 show that μ-Relief, ANOVA, SURF*, MultiSURF and MultiSURF* selected the three most relevant features. ReliefF, MI and Chi-Square selected two features of relevance, CMIM selected only one relevant feature whereas MRMR selected no relevant features. Therefore, μ-Relief has the ability to select the most relevant features comparable to existing techniques. As shown in Fig. 2, the estimated weights were observed against the number of neighbors k using μ-Relief. The difference between the weight estimates of relevant features and features of less or no relevance is evidently clear, even for 100 neighbors. Therefore, it can be concluded that the proposed algorithm is not short-sighted.

4.1.3 Experiment 3

To explain the neighbor selection behavior of μ-Relief, a synthetic dataset $E_3 = (I, F, T)$ as shown in Table 3 with a feature set defined as $F = \{A_1, A_2, A_3\}$ was used. The target binary class T is

$$T = A_3 \tag{23}$$

The differences between the selected neighbors using the ReliefF and μ-Relief methods are evident in Fig. 3. All hit and miss class instances are plotted in the 3-dimensional feature space of F using circles and triangles, respectively. The following assumptions were made to run the ReliefF and μ-Relief algorithm on E_3.

- 1 iteration ($m = 1$) was used to compute the weights,
- different random samples - I_1, I_6, I_{20} and I_{35} - were used for four distinct comparisons, and
- 3 neighbors ($k = 3$) were considered for weight estimation.

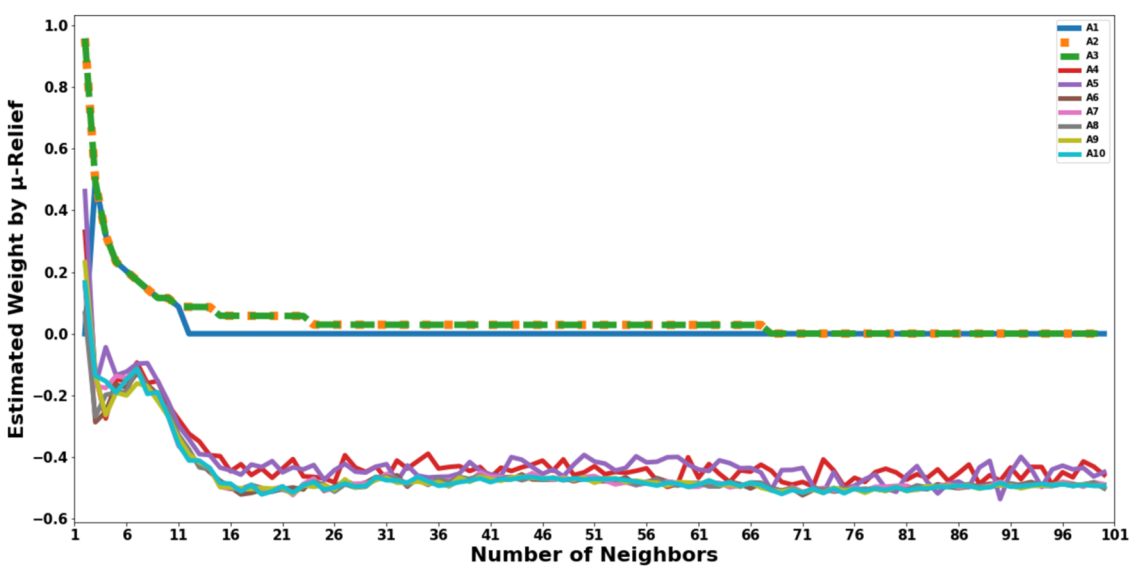

Fig. 2 Estimated Weights versus Number of Neighbors using μ-Relief

Table 3 Synthetic Dataset E_3

Instance	A_1	A_2	A_3	T
I_1	1	1	0	0
I_2	2	1	0	0
I_3	3	1	0	0
I_4	4	1	0	0
I_5	5	1	0	0
I_6	6	1	0	0
I_7	1	2	0	0
I_8	2	2	0	0
I_9	3	2	0	0
I_{10}	4	2	0	0
I_{11}	5	2	0	0
I_{12}	6	2	0	0
I_{13}	1	3	0	0
I_{14}	2	3	0	0
I_{15}	3	3	0	0
I_{16}	4	3	0	0
I_{17}	5	3	0	0
I_{18}	6	3	0	0
I_{19}	1	1	1	1
I_{20}	2	1	1	1
I_{21}	3	1	1	1
I_{22}	4	1	1	1
I_{23}	5	1	1	1
I_{24}	6	1	1	1
I_{25}	1	2	1	1
I_{26}	2	2	1	1
I_{27}	3	2	1	1
I_{28}	4	2	1	1
I_{29}	5	2	1	1
I_{30}	6	2	1	1
I_{31}	1	3	1	1
I_{32}	2	3	1	1
I_{33}	3	3	1	1
I_{34}	4	3	1	1
I_{35}	5	3	1	1
I_{36}	6	3	1	1

Fig. 3 shows that even in a small dataset, such as E_3, distinct sets of instances are selected for weight calculation by both methods.

4.2 Experiments on real-world datasets

Multi-domain real-world datasets from the UCI repository [41] and the literature [42] have been considered for the experiments described in this section. The selected datasets have nominal, continuous, and mixed features with binary and multi-class labels. The datasets have instances varying from 32 to 2162 and the number of features from 13 to 22283, as shown in Table 4. As each dataset had a different dimensionality, a limit was set to compare the feature selection algorithms. 30%, 40%, and 50% of the most relevant features determined by each feature selection method were considered for this purpose. Each dataset was split into ten folds, of which nine were used for feature selection and training the model and one for testing. SVM (Support Vector Machine), LR (Logistic Regression), XGBRF and RF (Random Forest) algorithms were used for classification. The averages of classification accuracies and standard deviations of the ten folds for the four classifiers have been reported in Tables 5, 6, 7 and 8. The number of iterations in μ-Relief is equal to the number of instances for each dataset, the same as in the *sk-rebate* implementation of RBAs.

Differences in neighbor selection Since the considerations to select neighbors in μ-Relief and ReliefF are different, the contrast in the chosen neighbors needs to be observed. Figure 4 shows the difference in the selected neighbors with μ-Relief and ReliefF using four datasets. The height of the bars in the figure shows the number of instances in the respective dataset. The random and the selected instances from the hit and miss(s) sets are marked differently. Figure 4(a) and (b) show the comparison of the selected instances of the same and different classes with the two algorithms when the *Heart-spectf* dataset was used and the query instances were I_{100} and I_{225}, respectively. The distributions of the selected instances for the hit and miss sets in Fig. 4(c) and (d) reveal that ReliefF was more restricted in certain localities as compared to μ-Relief while choosing neighbors from the *leukemia* dataset. μ-Relief showed a more dataset-wide and distributed coverage than ReliefF to obtain informative instances from the *colon* dataset, as depicted in Fig. 4(e) and (f) when the query points were I_{25} and I_{50}. Similarly, a clear difference in the selected instances for the hit set and multiple miss sets was observed when the two algorithms were applied to the *wine* dataset (Fig. 4(g) and (h)).

4.2.1 Paired two-tailed T-test

The paired two-tailed t-test, based on Tables 5, 6, 7 and 8, was used to evaluate the performance of μ-Relief based on mean accuracy. In this test, the proposed method is compared with other feature selection algorithms and the "+" indicates if it performs better (win), "-" for worse (loss), and "=" for equivalent performance (tie) at a statistical significance level of 0.05. The highest mean accuracy is shown in bold in each row. When SVM was used for classification, the average accuracy obtained in μ-Relief for 30%, 40% and 50% feature sets were the highest for eight datasets, as shown in Table 5. It is also visible from Table 5 that μ-Relief performs the best on 24

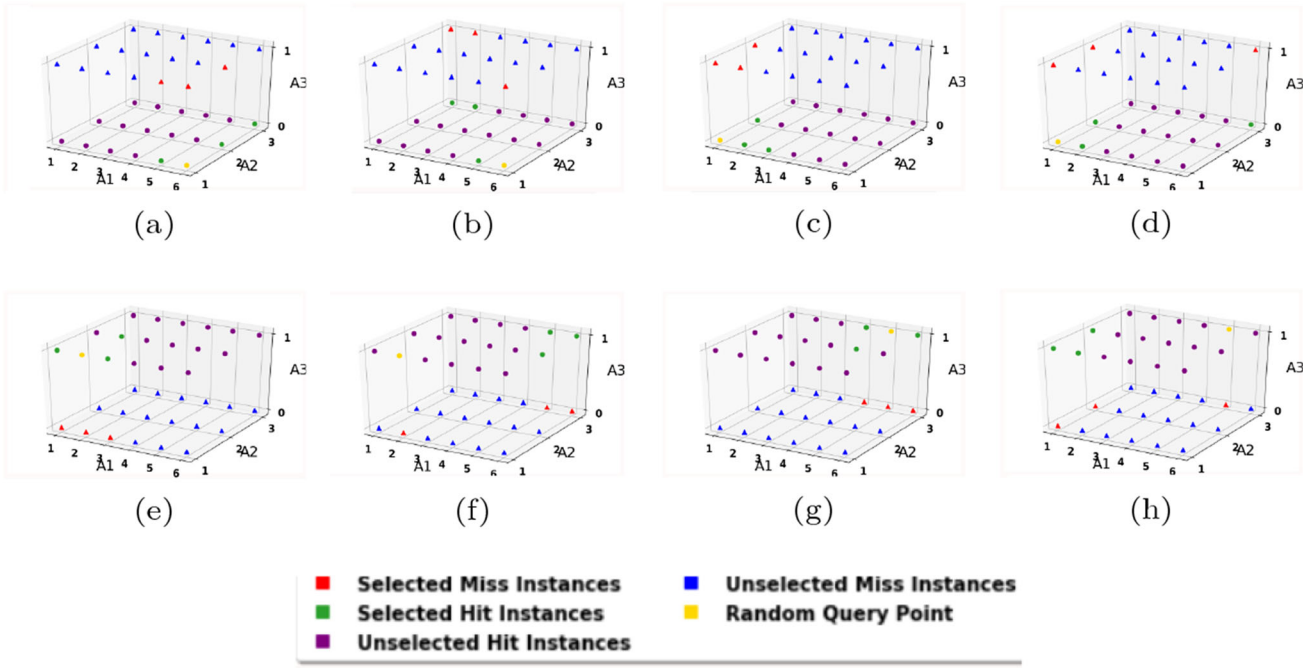

Fig. 3 Scatter plots of Selected Neighbors using ReliefF and μ-Relief in Experiment 3. Random query point in (**a**), (**b**) is I_6, in (**c**), (**d**) is I_1, in (**e**), (**f**) is I_{20}, and in (**g**), (**h**) I_{35}

of 39 subsets (30%, 40%, and 50% top-ranked feature set of 13 datasets). It outperforms the second-best feature selection technique (ANOVA) by about 26%. All the feature selectors obtained the same accuracy with the *GLI*_85 dataset. On the *Parkinsons* dataset, for the 30% set, MultiSURF and μ-Relief yielded the best and the second-best accuracies, respectively, while for the 40% and 50% feature sets, MRMR performed best.

Results in Table 6 show that with the LR classifier, μ-Relief consistently performs well on 23 out of 39 subsets, which is an approximately 21% better performance than the second next-best feature selection method (ANOVA). μ-Relief achieved the highest mean accuracy with 30% features for 7 datasets, 40% features for 9 datasets, and 50% features for 4 datasets with LR. All feature selection techniques obtained the same mean accuracy using 40% and 50% features from the *leukemia* dataset. Table 7 shows that using the μ-Relief method with the XGBRF classifier recorded the best performance in 17 out of 39 subsets. μ-Relief achieves the highest accuracy on 6 datasets for both 30% and 50%

Table 4 Real-world Datasets used in the study

Dataset	Instances	Features	Classes	30% features	40% features	50% features	Abbreviation	source
ALLAML	72	7129	2	2139	2852	3565	AM	[42]
Arrhythmia	452	279	13	84	112	139	AR	[42]
colon	62	2000	2	600	800	1000	CO	[42]
fetal	2126	21	3	6	9	11	FE	[41]
heartSpectf	267	44	2	14	18	22	HS	[41]
Lung Cancer	32	57	3	17	22	27	LC	[41]
lymphoma	96	4027	9	1208	1611	2013	LY	[42]
leukemia	72	7070	2	2121	2828	3535	LE	[42]
Musk1	476	166	2	50	67	83	MU	[41]
Parkinsons	195	22	2	6	9	11	PA	[41]
Prostate	102	5967	2	1790	2387	2983	PR	[42]
Wine	178	13	3	4	6	7	WI	[41]
GLI_85	85	22283	2	6685	8914	11142	GL	[42]

Table 5 Comparison of classification accuracies (%) (mean ±standard deviation) on the different datasets with SVM

Dataset	Top (%) features	Chi-square	ANOVA	MI	MRMR	CMIM	MultiSURF	SURF*	MultiSURF*	ReliefF	μ-Relief
PA	30	80.42±8.38(+)	80.42±8.38(+)	82.44±10.2(+)	74.81±4.58(+)	77.81±9.70(+)	**86.00±8.91**(-)	69.63±8.22(+)	71.71±7.04(+)	84.36±12.6(+)	85.50± 10.6
	40	75.39±1.86(=)	75.39±1.86(=)	75.39±1.86(=)	**76.97±8.01**(-)	75.39±1.86(=)	75.39±1.86(=)	75.39±1.86(=)	75.39±1.86(=)	75.39±1.86(=)	75.39±1.86
	50	75.39±1.86(=)	75.39±1.86(=)	75.39±1.86(=)	**76.97±8.01**(-)	75.39±1.86(=)	75.39±1.86(=)	75.39±1.86(=)	75.39±1.86(=)	75.39±1.86(=)	75.39±1.86
MU	30	70.85±9.51(+)	74.60±7.55(+)	76.50±9.09(-)	76.31±9.12(-)	**76.95±10.15**(-)	72.54±10.27(+)	70.66±12.83(+)	71.70±9.64(+)	68.75±11.59(+)	75.23±7.73
	40	69.51±0.66(+)	73.28±6.79(-)	76.50±9.09(-)	75.87±2.67(-)	**76.95±10.15**(-)	69.51±0.66(+)	69.51±0.66(+)	69.51±0.66(+)	69.51±0.66(+)	72.54±10.26
	50	69.51±0.66(+)	72.83±5.41(-)	76.50±9.02(-)	75.87±2.67(-)	**76.95±10.15**(-)	69.51±0.66(+)	69.51±0.66(+)	69.51±0.66(+)	69.51±0.66(+)	72.51±10.26
FE	30	84.89±9.51(-)	84.37±8.39(-)	84.75±8.39(-)	80.76±2.20(-)	77.84±0.39(+)	82.43±12.16(-)	80.55±14.68(-)	82.20±15.06(-)	**84.98±10.14**(-)	77.94±1.19
	40	80.93±10.31(=)	**82.53±10.37**(-)	80.18±12.41(+)	78.64±6.57(+)	79.65±12.16(+)	79.42±13.53(+)	76.17±13.98(+)	75.52±13.68(+)	81.17±11.09(-)	80.93±12.81
	50	80.93±10.31(-)	**82.53±10.37**(-)	80.18±12.41(-)	78.64±6.57(-)	79.65±12.16(-)	80.04±13.22(-)	75.56±14.06(+)	75.56±14.06(+)	81.17±11.09(-)	77.78±10.55
WI	30	70.88±7.09(+)	70.88±7.09(+)	70.88±7.09(+)	87.77±6.87(+)	78.59±5.58(+)	90.52±8.61(+)	87.48±6.48(+)	79.28±6.34(+)	92.25±7.10(+)	**93.30±6.93**
	40	64.67±10.04(+)	64.67±10.04(+)	64.67±10.04(+)	88.26±8.76(+)	84.90±8.19(+)	64.44±5.70(+)	65.03±6.28(+)	65.03±6.28(+)	64.67±10.04(+)	**91.63±7.12**
	50	64.67±10.04(+)	64.67±10.04(+)	64.67±10.04(+)	88.26±8.76(+)	84.90±8.19(+)	63.88±5.94(+)	64.47±7.00(+)	64.47±7.00(+)	64.67±10.04(+)	**92.18±7.10**
LC	30	52.50±3.76(+)	60.00±2.17(+)	44.17±2.17(+)	33.33±2.71(+)	47.50±2.23(+)	55.83±3.07(+)	45.83±2.21(+)	43.33±2.01(+)	46.67±2.79(+)	**61.10±2.38**
	40	**54.05±7.95**(=)	**54.05±7.95**(=)	**54.05±7.95**(=)	44.50±5.11(+)	**54.05±7.95**(=)	50.00±4.05(+)	50.00±4.05(+)	50.00±4.05(+)	51.66±6.67(+)	**54.05±7.95**
	50	**54.05±7.95**(=)	**54.05±7.95**(=)	**54.05±7.95**(=)	44.50±5.11(+)	**54.05±7.95**(=)	50.00±4.05(+)	50.00±4.05(+)	50.00±4.05(+)	51.66±6.67(+)	**54.05±7.95**
PR	30	92.09±8.70(+)	92.09±8.70(+)	**93.09±8.97**(-)	92.09±8.70(+)	86.27±7.92(+)	90.18±8.95(+)	92.09±7.46(=)	90.18±6.33(+)	90.18±8.95(+)	92.09±7.42
	40	**92.09±8.70**(=)	**92.09±8.70**(=)	**92.09±8.70**(=)	**92.09±8.70**(=)	91.18±8.29(+)	**92.09±8.70**(=)	**92.09±8.70**(=)	**92.09±8.70**(=)	**92.09±8.70**(=)	**92.09±8.70**
	50	**92.09±8.70**(-)	**92.09±8.70**(-)	**92.09±8.70**(-)	**92.09±8.70**(-)	91.18±8.29(-)	90.18±9.42(+)	91.09±8.95(-)	90.18±9.42(+)	90.18±9.42(+)	91.09±8.95
LY	30	**87.11±8.45**(=)	**87.11±8.45**(=)	85.12±7.47(+)	81.12±7.79(+)	78.00±11.63(+)	81.00±9.39(+)	76.78±11.75(+)	76.77±13.69(+)	83.12±8.50(+)	**87.11±8.45**
	40	**96.61±3.11**(=)	**96.61±3.11**(=)	94.50±6.38(+)	93.44±4.43(+)	93.44±4.43(+)	95.65±4.89(+)	95.52±3.27(+)	93.65±5.86(+)	94.50±6.38(+)	**96.61±3.11**
	50	**96.61±3.11**(=)	**96.61±3.11**(=)	94.50±6.38(+)	93.44±4.43(+)	93.44±4.43(+)	85.21±3.47(+)	84.11±3.42(+)	84.11±3.42(+)	84.11±3.42(+)	**96.61±3.11**
AR	30	67.54±5.83(+)	71.11±5.61(+)	72.80±5.70(+)	67.30±5.51(+)	63.95±3.81(+)	72.56±7.20(+)	68.49±6.36(+)	69.93±6.63(+)	71.11±3.65(+)	**73.28±3.46**
	40	64.18±5.31(+)	59.43±2.75(+)	57.99±2.13(+)	61.10±3.93(+)	59.43±2.75(+)	65.40±4.33(+)	65.87±4.76(+)	64.68±5.79(+)	62.75±3.96(+)	**67.06±4.87**
	50	64.66±5.03(+)	59.43±2.75(+)	57.99±2.13(+)	61.10±3.93(+)	59.43±2.75(+)	64.44±3.04(+)	60.38±3.20(+)	60.62±2.76(+)	64.68±3.41(+)	**68.49±5.44**
CO	30	85.00±15.71(+)	83.5±14.92(+)	83.80±16.24(+)	81.92±18.96(+)	83.80±14.43(+)	83.80±16.24(+)	84.2±13.83(+)	83.78±16.24(+)	80.4±16.00(+)	**85.2±15.65**
	40	81.57±14.93(+)	81.57±14.93(+)	80.71±14.22(+)	72.61±14.29(+)	74.28±19.39(+)	80.37±15.22(+)	80.92±14.49(+)	80.92±14.49(+)	80.92±13.76(+)	**83.5±12.36**
	50	81.57±14.93(+)	81.57±14.93(+)	80.71±14.22(+)	72.61±14.29(+)	74.28±19.39(+)	81.80±12.94(+)	82.47±11.63(+)	80.71±14.22(+)	82.14±13.93(+)	**82.61±14.22**
LE	30	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	87.32±14.90(+)	93.03±9.46(+)	**98.57±4.28**(=)	93.03±9.46(+)	93.03±9.46(+)	97.14±8.57(+)	**98.57±4.28**
	40	**98.57±4.28**(=)	**98.57±4.28**(=)	97.14±5.71(+)	94.46±9.32(+)	93.03±9.46(+)	**98.57±4.28**(=)	**98.57±4.28**(=)	97.14±5.71(+)	**98.57±4.28**(=)	**98.57±4.28**
	50	**98.57±4.28**(=)	**98.57±4.28**(=)	97.14±5.71(+)	94.46±9.32(+)	93.03±9.46(+)	**98.57±4.28**(=)	97.14±5.71(+)	95.71±6.54(+)	**98.57±4.28**(=)	**98.57±4.28**

Table 5 continued

Dataset	Top (%) features	Chi-square	ANOVA	MI	MRMR	CMIM	MultiSURF	SURF*	MultiSURF*	ReliefF	μ-Relief
HS	30	79.41±1.65(+)	79.04±2.27(+)	79.04±3.66(+)	79.78±2.34(+)	79.41±1.65(+)	79.07±3.92(+)	79.41±1.65(+)	78.25±3.13(+)	79.80±2.26(+)	**79.95±2.18**
	40	78.66±2.27(+)	79.03±1.74(+)	79.78±1.66(+)	79.04±2.27(+)	79.40±2.97(+)	77.52±7.94(+)	77.09±8.74(+)	77.12±6.76(+)	78.21±10.85(+)	**79.95±2.18**
	50	78.66±2.27(+)	79.03±1.74(+)	**79.78±1.66**	79.04±2.27(+)	79.41±1.65(+)	79.37±5.71(+)	79.78±2.34(+)	78.24±4.87(+)	77.49±5.55(+)	**79.78±1.66**
GL	30	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58
	40	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58
	50	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58(=)	69.58±4.58
AM	30	94.10±7.85(+)	94.10±7.85(+)	93.45±6.67(+)	94.10±7.85(+)	90.17±12.80(+)	94.46±6.79(+)	90.17±11.10(+)	91.60±11.35(+)	**95.89±8.96**(=)	**95.89±6.29**
	40	**95.89±6.29**(-)	**95.89±6.29**(-)	**95.89±6.29**(-)	91.03±7.41(+)	93.89±2.69(+)	94.14±5.71(+)	89.42±11.42(+)	89.42±11.42(+)	94.32±5.37(=)	94.32±5.37
	50	**95.89±6.29**(=)	**95.89±6.29**(=)	**95.89±6.29**(=)	91.03±7.41(+)	93.89±2.69(+)	94.14±5.71(+)	90.00±14.35(+)	90.00±14.35(+)	88.69±13.28(+)	**95.89±6.29**

Table 6 Comparison of classification accuracies (%) (mean ± standard deviation) on the different datasets with LR

Dataset	Top (%) features	Chi-square	ANOVA	MI	MRMR	CMIM	MultiSURF	SURF*	MultiSURF*	ReliefF	μ-Relief
PA	30	81.31±11.5(+)	80.81±10.8(+)	83.36±11.0(+)	82.86±10.3(+)	79.84±9.93(+)	79.76±11.21(+)	69.50±12.44(+)	72.57±12.26(+)	79.28±10.1(+)	**83.94±9.05**
	40	81.81±11.81(+)	81.31±11.12(+)	81.28±12.27(+)	82.84±12.20(+)	79.31±12.22(+)	81.31±11.12(+)	75.39±1.86(+)	75.39±1.86(+)	82.94±10.09(+)	**83.36±11.67**
	50	81.81±11.81(+)	81.31±11.12(+)	81.28±12.27(+)	81.31±11.12(+)	79.31±12.22(+)	81.31±11.12(+)	75.39±1.86(+)	75.39±1.86(+)	80.78±11.58(+)	**82.84±12.20**
MU	30	67.23±7.18(+)	70.77±7.50(+)	62.17±9.29(+)	69.96±6.37(+)	68.68±5.91(+)	68.52±8.57(+)	67.71±13.38(+)	69.38±11.79(+)	69.55±6.78(+)	**73.36±8.07**
	40	75.26±10.18(+)	75.50±11.57(+)	76.32±9.65(+)	75.70±10.77(+)	72.34±12.88(+)	75.68±8.70(+)	74.42±10.46(+)	75.77±6.67(+)	76.23±10.44(+)	**76.38±8.45**
	50	75.06±10.49(+)	75.69±1031(+)	**77.17±10.16**(-)	76.12±10.68(+)	76.95±8.56(+)	76.52±9.90(+)	75.87±6.92(+)	75.24±7.16(+)	76.32±8.90(+)	77.16±9.95
FE	30	82.40±7.64(-)	83.01±7.50(-)	**84.28±7.55**(-)	79.40±3.66(+)	77.19±2.95(+)	80.94±6.49(+)	80.93±11.08(+)	80.36±11.15(+)	82.96±7.17(-)	81.94±1.46
	40	84.89±5.45(+)	84.57±7.56(+)	84.75±8.72(+)	81.42±4.19(+)	80.70±8.55(+)	84.17±8.91(+)	81.17±5.70(+)	81.60±5.60(+)	84.94±8.70(+)	**84.98±7.53**
	50	84.89±5.45(+)	84.57±7.56(+)	84.75±8.72(+)	81.42±4.19(+)	80.70±8.55(+)	**85.50±8.75**(-)	80.08±9.35(+)	80.84±6.59(+)	85.08±8.72(-)	84.98±7.53
WI	30	91.63±6.67(+)	91.63±6.67(+)	**94.41±5.55**(-)	87.77±6.87(+)	85.51±5.58(+)	90.52±8.61(+)	87.48±6.48(+)	85.26±6.34(+)	92.25±7.10(+)	93.26±5.44
	40	**96.11±0.35**(=)	**96.11±0.35**(=)	**96.11±0.35**(=)	91.63±0.56(+)	93.85±0.57(+)	93.88±0.76(+)	88.85±0.96(+)	89.41±0.94(+)	**96.11±0.35**(=)	**96.11±0.35**
	50	**96.11±0.35**(-)	**96.11±0.35**(-)	**96.11±0.35**(-)	91.63±0.56(+)	93.85±0.57(+)	93.34±8.88(+)	91.63±7.54(+)	92.18±7.10(+)	96.07±2.56(-)	95.52±0.41
LC	30	45.83±13.44(+)	55.50±12.76(+)	55.50±12.76(+)	36.67±12.89(+)	42.50±12.59(+)	53.33±12.76(+)	55.50±12.76(+)	43.33±12.01(+)	43.33±12.73(+)	**57.50±12.59**
	40	66.67±13.43(+)	66.67±13.57(+)	63.33±13.87(+)	48.33±12.62(+)	56.66±13.45(+)	62.50±13.56(+)	36.67±12.66(+)	48.33±13.02(+)	**68.33±13.20**	**68.33±13.20**
	50	66.67±13.43(+)	**69.16±13.57**(-)	63.33±13.87	48.33±12.62	56.66±13.45	46.66±12.79	33.34±12.47	36.67±12.89	65.83±13.38	68.33±13.20
PR	30	92.09±8.70(+)	92.09±870(+)	93.01±9.00(+)	92.09±8.70(+)	89.27±6.80(+)	90.18±8.95(+)	91.09±9.42(+)	90.18±8.95(+)	91.09±9.42(+)	**93.09±8.97**
	40	92.09±8.70(+)	**93.00±9.00**(=)	91.09±9.42(+)	92.09±8.70(+)	89.27±6.80(+)	**93.00±9.00**(=)	92.09±8.70(+)	**93.00±9.00**(=)	92.09±8.70(+)	**93.00±9.00**
	50	92.09±8.70(-)	**93.00±9.00**(-)	91.09±9.42(=)	**93.00±9.00**(-)	89.27±6.80(+)	91.18±6.98(-)	92.09±8.70(-)	91.18±6.98(-)	91.09±9.42(=)	91.09±9.42
LY	30	95.67±7.27(=)	**96.67±7.11**(-)	95.56±8.89(+)	91.33±10.85(+)	92.44±11.97(+)	91.22±10.86(+)	91.22±10.86(+)	93.44±10.14(+)	89.11±13.15(+)	95.67±7.27
	40	92.33±8.49(+)	94.45±5.18(+)	**96.67±7.11**(-)	90.22±7.27(+)	92.33±7.07(+)	95.56±5.18(-)	95.56±5.18(-)	95.56±5.18(-)	93.44±6.30(+)	95.44±3.45
	50	92.33±8.49(+)	94.45±5.18(+)	96.67±7.11(+)	90.22±7.27(+)	92.33±7.07(+)	93.13±6.33(+)	93.13±6.33(+)	92.33±7.13(+)	92.33±7.07(+)	**96.67±.3.28**
AR	30	61.09±5.63(+)	53.21±6.56(+)	56.33±7.91(+)	54.87±5.59(+)	55.59±5.78(+)	68.97±8.69(+)	67.05±6.14(+)	67.78±6.85(+)	68.97±7.5(+)1	**68.97±4.91**
	40	63.24±8.32(+)	56.09±6.18(+)	63.48±5.83(+)	59.90±5.31(+)	56.09±6.18(+)	64.44±7.21(+)	64.66±7.58(+)	64.92±6.94(+)	60.85±7.46(+)	**70.41±6.63**
	50	63.24±8.32(+)	56.09±6.18(+)	63.48±5.83(+)	59.90±5.31(+)	56.09±6.18(+)	71.60±8.33(-)	70.14±6.63(-)	71.12±5.65(-)	**73.27±7.71**(-)	68.26±5.90
CO	30	**85.47±13.81**(=)	**85.47±13.81**(=)	83.80±14.93(+)	82.14±15.32(+)	82.30±13.43(+)	83.80±12.94(+)	82.14±15.32(+)	83.80±12.94(+)	83.80±12.94(+)	**85.47±13.81**
	40	82.38±15.36(+)	85.14±14.45(-)	83.80±12.94(-)	77.38±16.87(+)	79.52±15.32(+)	**85.47±13.81**(-)	85.14±14.45(-)	83.80±12.94(-)	**85.47±13.81**(-)	82.61±14.90
	50	82.38±15.36(+)	**87.14±14.45**(-)	83.80±12.94(+)	77.38±16.87(+)	79.52±15.32(+)	84.01±14.44(+)	84.40±14.87(-)	85.47±13.81(-)	85.47±13.81(-)	84.28±12.80
LE	30	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	97.14±5.71(+)	97.14±5.71(+)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**
	40	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**
	50	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**(=)	**98.57±4.28**

Table 6 continued

Dataset	Top (%) features	Chi-square	ANOVA	MI	MRMR	CMIM	MultiSURF	SURF*	MultiSURF*	ReliefF	μ-Relief
HS	30	75.66±5.28(+)	76.76±6.96(=)	77.53±6.53(-)	74.20±7.09(+)	75.69±4.25(+)	**80.17±2.20**(-)	79.41±1.65(-)	79.41±1.65(-)	78.68±3.53(-)	76.76±6.96
	40	78.94±10.07(+)	**81.56±9.51**(-)	81.18±10.64(-)	76.77±6.47(+)	76.75±7.09(+)	80.52±2.77(-)	79.41±1.65(+)	79.78±2.34(-)	79.38±2.62(+)	79.75±5.64
	50	78.94±10.07(+)	**81.56±9.51**(-)	81.18±10.64(+)	76.77±6.47(+)	76.75±7.01(+)	79.74±4.65(+)	80.52±3.62(+)	79.75±3.89(+)	76.41±4.04(+)	81.18±10.64
GL	30	86.94±8.49(+)	85.83±7.37(+)	87.08±8.12(+)	85.69±7.32(+)	86.94±11.03(+)	89.16±10.03(+)	90.42±10.50(+)	90.42±10.51(+)	91.52±9.67(+)	**91.52±8.67**
	40	87.08±8.12(+)	84.72±10.39(+)	87.08±9.52(+)	84.72±10.39(+)	87.08±8.12(+)	85.83±10.80(+)	88.19±7.51(+)	88.19±7.51(+)	86.94±9.84(+)	**88.67±10.85**
	50	87.08±8.12(+)	84.72±10.39(+)	87.08±9.52(+)	84.72±10.39(+)	87.08±8.12(+)	87.08±9.52(+)	87.08±9.52(+)	88.19±7.51(+)	84.72±10.3(+)	**88.67±10.85**
AM	30	93.21±6.25(=)	93.21±6.25(=)	90.15±7.89(+)	91.23±7.53(+)	90.25±11.36(+)	**94.46±6.79**(-)	90.17±11.10(+)	91.48±10.28(+)	93.21±6.25(=)	93.21±6.25
	40	**98.73±2.85**(=)	**98.73±2.85**(=)	**98.73±2.85**(=)	95.71±9.14(+)	97.41±5.14(+)	**98.73±2.85**(=)	91.48±11.42(+)	92.86±9.58(+)	**98.73±2.85**(=)	**98.73±2.85**
	50	**98.73±2.85**(=)	**98.73±2.85**(=)	**98.73±2.85**(=)	95.71±9.14(+)	97.41±5.14(+)	**98.73±2.85**(=)	91.48±11.42(+)	92.86±9.58(+)	94.28±8.32(+)	**98.73±2.85**

Table 7 Comparison of classification accuracies (%) (mean ± standard deviation) on the different datasets with XGBRF

Dataset	Top (%) features	Chi-square	ANOVA	MI	MRMR	CMIM	MultiSURF	SURF*	MultiSURF*	ReliefF	μ-Relief
PA	30	**84.94±9.02**(-)	83.36±10.7(-)	82.86±10.0(=)	84.89±10.7(-)	79.86±12.2(+)	82.94±9.85(-)	72.23±8.14(+)	75.23±9.56(+)	84.92±10.1(-)	82.86±12.30
	40	**83.89±11.46**(-)	82.97±9.26(-)	81.92±10.37(+)	81.39±10.48(+)	79.34±12.11(+)	81.42±9.07(+)	74.73±12.70(+)	76.28±12.08(+)	84.00±7.85(-)	82.86±11.05
	50	**83.89±11.46**(-)	82.97±9.26(-)	81.92±10.37(+)	81.39±10.48(+)	79.34±12.11(+)	81.42±9.61(+)	74.76±12.84(+)	74.10±14.54(+)	81.94±8.43(+)	82.89±10.34
MU	30	**69.79±10.66**(-)	68.53±10.73(-)	63.22±9.57(+)	65.56±8.62(+)	63.50±11.26(+)	68.33±10.81(-)	65.18±9.71(+)	62.88±11.08(+)	68.13±10.34(-)	68.10±9.30
	40	**68.52±10.68**(-)	67.69±10.07(+)	65.79±8.52(+)	66.01±7.45(+)	63.90±12.63(+)	67.69±10.01(+)	66.41±8.85(+)	67.47±8.93(+)	67.69±10.52(+)	68.46±11.70
	50	**68.52±10.68**(-)	67.69±10.07(-)	65.79±8.52(-)	66.01±7.45(-)	63.90±12.63(+)	67.48±9.00(-)	67.04±10.09(-)	67.46±8.13(-)	66.21±9.44(-)	65.12±11.61
FE	30	**89.64±8.81**(-)	88.65±7.19(+)	81.04±3.18(+)	78.40±5.64(+)	80.52±4.97(+)	85.40±8.38(+)	84.97±9.71(+)	85.77±9.70(+)	89.03±7.88(-)	88.98±7.99
	40	88.89±7.72(-)	**89.17±6.94**(-)	88.51±7.22(+)	83.39±3.95(+)	88.32±8.01(-)	88.79±7.71(-)	85.35±8.81(+)	85.49±8.74(+)	88.46±7.80(-)	87.81±10.26
	50	88.89±7.72(-)	**89.17±6.94**(-)	88.51±7.22(-)	83.39±3.95(-)	88.32±8.01(-)	88.70±7.14(-)	85.50±8.04(-)	85.45±8.30(-)	88.51±7.99(-)	82.53±9.71
WI	30	91.11±8.31(+)	91.30± 6.07(+)	**95.00±3.55**(-)	92.70±5.54(+)	89.80±6.63(+)	89.93±6.43(+)	88.13±7.80(+)	87.75±5.44(+)	91.11±8.31(+)	94.41±5.55
	40	96.11±0.35(+)	96.11±0.35(+)	96.11±0.35(+)	96.63±0.51(+)	93.82±0.46(+)	96.10±0.55(+)	92.15±0.57(+)	91.60±0.62(+)	95.00±0.67(+)	**97.22±0.37**
	50	96.11±0.65(+)	96.11±0.50(+)	96.11±0.50(+)	96.63±0.51(+)	93.82±0.46(+)	96.11±0.55(+)	93.82±0.63(+)	93.26±0.74(+)	96.11±0.65(+)	**97.22±0.37**
LC	30	53.70±12.51(+)	57.40±12.76(+)	59.25±12.89(+)	50.01±11.61(+)	42.50±12.59(+)	53.70±12.78(+)	41.66±12.03(+)	37.96±12.39(+)	49.07±12.46(+)	**62.03±12.66**
	40	53.33±13.25(+)	60.01±13.37(+)	60.05±13.68(+)	46.67±13.16(+)	57.50±13.67(+)	59.16±13.66(+)	35.83±12.73(+)	45.02±12.58(+)	56.67±13.11(+)	**60.08±12.62**
	50	**63.34±12.35**(=)	60.01±13.37(+)	60.05±13.68(+)	46.67±13.16(+)	57.50±13.67(+)	55.83±14.01(+)	35.83±12.73(+)	42.50±12.12(+)	**63.34±12.35**(=)	**63.34±12.35**
	30	90.09±8.94(+)	**91.00±9.43**(=)	**91.00±9.43**(=)	90.27±7.53(+)	82.45±11.07(+)	90.00±10.00(+)	89.03±9.43(+)	**91.00±9.43**(=)	**91.00±9.43**(=)	**91.00±9.43**
	40	88.09±9.82(+)	89.00±10.44(=)	89.00±10.44(=)	**90.18±6.33**(-)	89.27±8.13(-)	89.00±10.44(=)	88.09±9.82(+)	88.09±9.82(+)	89.00±10.44(=)	89.00±10.44
	50	88.09±9.82(=)	89.00±10.44(-)	89.00±10.44(-)	**90.18±6.33**(-)	89.27±8.13(-)	89.00±10.44(-)	87.00±10.05(+)	89.00±10.44(-)	89.00±10.44(-)	88.09±9.82
LY	30	83.22±10.44(=)	**85.11±12.48**(-)	83.00±13.42(+)	77.78±15.58(+)	78.89±13.41(+)	83.11±13.65(+)	78.78±16.34(+)	79.00±12.97(+)	77.78±14.45(+)	83.22±9.41
	40	86.11±14.37(+)	85.11±12.48(+)	85.00±14.29(+)	81.01±11.32(+)	76.78±14.40(+)	86.22±10.74(+)	79.89±15.05(+)	77.89±15.21(+)	80.05±13.38(+)	**86.33±11.80**
	50	86.33±14.37(+)	85.11±12.48(+)	85.00±14.29(+)	81.01±11.32(+)	76.78±14.40(+)	81.11±8.98(+)	81.89±16.96(+)	79.02±13.72(+)	81.11±12.67(+)	**86.33±11.80**
AR	30	69.21±4.15(+)	71.83±5.62(+)	74.41±6.04(+)	70.18±5.28(+)	67.06±3.82(+)	73.01±6.44(+)	69.70±4.95(+)	69.93±4.97(+)	71.84±6.51(+)	**74.69±5.69**
	40	72.07±6.03(+)	70.88±4.96(+)	**75.41±5.24**(-)	70.65±6.27(+)	70.87±4.89(+)	73.73±6.32(+)	69.68±4.15(+)	71.59±5.27(+)	74.45±6.06(=)	74.45±5.01
	50	72.07±6.03(+)	70.88±4.96(+)	**75.41±5.24**(-)	70.65±6.27(+)	70.87±4.89(+)	74.69±4.95(+)	69.93±5.40(+)	71.13±6.21(+)	73.74±4.99(+)	75.40±6.17
CO	30	80.10±14.52(+)	83.33±12.90(+)	81.66±17.40(+)	77.14±18.75(+)	83.57±10.56(+)	80.00±12.47(+)	78.57±15.09(+)	80.02±16.32(+)	78.33±15.00(+)	**85.00±13.72**
	40	76.67±16.99(+)	83.33±16.67(+)	81.67±15.72(+)	76.90±17.10(+)	80.23±14.60(+)	83.33±12.90(+)	83.33±14.91(+)	78.33±16.74(+)	83.33±16.67(+)	**83.33±12.89**
	50	76.67±16.99(+)	**83.33±16.67**(=)	81.67±15.72(+)	76.90±17.10(+)	80.23±14.60(+)	81.67±15.72(+)	80.01±16.32(+)	78.33±16.74(+)	**83.33±16.67**(=)	**83.33±16.67**
LE	30	92.58±7.14(+)	97.14±5.71(+)	97.14±5.71(+)	88.75±10.65(+)	90.17±12.80(+)	**98.57±4.28**(=)	91.42±9.47(+)	91.78±10.83(+)	97.14±5.71(+)	**98.57±4.28**
	40	92.85±7.14(+)	97.14±5.71(+)	97.14±5.71(+)	92.85±7.14(+)	94.28±6.99(+)	**98.57±4.28**(=)	94.28±6.99(+)	89.11±11.96(+)	**98.57±4.28**(=)	**98.57±4.28**
	50	92.85±7.14(+)	97.14±5.71(+)	97.14±5.71(+)	92.85±7.14(+)	94.28±6.99(+)	**98.57±4.28**(=)	94.28±6.99(+)	94.28±6.99(+)	**98.57±4.28**(=)	**98.57±4.28**

Table 7 continued

Dataset	Top (%) features	Chi-square	ANOVA	MI	MRMR	CMIM	MultiSURF	SURF*	MultiSURF*	ReliefF	μ-Relief
HS	30	76.82±9.56(+)	**80.52±7.45**(-)	80.15±8.15(-)	78.33±6.03(+)	77.92±3.38	79.43±4.95(-)	79.78±4.99(-)	77.54±6.34(+)	77.17±8.46(+)	78.71±5.58
	40	77.12±5.89(+)	81.63±4.31(-)	**83.14±5.13**(-)	78.30±3.49(+)	77.13±3.65(+)	77.90±5.73(+)	78.27±7.49(+)	75.99±4.68(+)	76.35±9.64(+)	80.52±7.45
	50	77.12±5.89(+)	81.63±4.31(-)	**83.14±5.13**(-)	78.30±3.49(+)	77.13±3.65(+)	79.00±5.51(+)	75.61±5.35(+)	77.38±6.38(+)	76.36±7.97(+)	81.28±4.64
GL	30	85.83±7.37(+)	87.91±7.93(+)	89.30±8.31(+)	88.33±7.06(+)	84.30±8.28(+)	87.91±7.93(+)	89.30±6.15(+)	**90.41±6.92**(-)	90.41±7.32(=)	90.41±7.32
	40	85.83±7.37(+)	87.91±7.93(+)	89.30±8.31(-)	88.33±7.06(+)	84.30±8.28(+)	85.56±10.74(+)	89.16±8.32(=)	**90.41±6.92**(-)	88.05±7.51(+)	89.16±8.32
	50	85.83±7.37(+)	87.91±7.93(+)	89.30±8.31(-)	88.33±7.06(=)	84.30±8.28(+)	85.56±10.74(+)	89.16±8.32(-)	**90.41±6.92**(-)	86.80±10.15	88.33±7.06
AM	30	**93.21±8.81**(=)	93.21±10.88(+)	93.21±10.88(+)	91.96±10.77(+)	90.53±8.37(+)	90.35±12.37(+)	86.07±12.79(+)	84.64±11.93(+)	91.78±10.83(+)	**93.21±10.88**
	40	**93.21±8.81**(-)	**93.21±8.81**(-)	**93.21±8.81**(-)	91.96±10.77(=)	91.96±10.77(=)	90.35±12.37(+)	87.50±13.45(+)	86.07±12.79(+)	90.35±12.37(+)	91.96±10.77
	50	**93.21±8.81**(=)	**93.21±8.81**(=)	**93.21±8.81**(=)	91.96±10.77(+)	91.96±10.77(+)	90.35±12.37(+)	87.50±13.45(+)	86.07±12.79(+)	90.35±12.37(+)	**93.21±8.81**

Table 8 Comparison of classification accuracies (%) (mean ± standard deviation) on the different datasets with RF

Dataset	Top (%) features	Chi-square	ANOVA	MI	MRMR	CMIM	MultiSURF	SURF*	MultiSURF*	ReliefF	μ-Relief
±PA	30	85.94±10.3(+)	83.39±10.2(+)	83.39±9.96(+)	**86.53±7.89**(-)	84.39±12.9(+)	85.97±9.72(=)	77.57±9.46(+)	72.63±10.7(+)	82.86±10.5(+)	85.97±8.19
	40	84.47±7.48(+)	84.47±9.91(+)	85.42±9.45(+)	83.81±11.97(+)	82.36±11.94(+)	86.05±8.36(+)	75.13±13.70(+)	74.10±13.75(+)	82.34±11.58(+)	**86.50±8.31**
	50	84.47±7.48(+)	83.94±10.05(+)	84.92±9.33(+)	80.34±11.57(+)	82.36±11.94(+)	86.05±8.36(+)	74.10±15.04(+)	74.60±15.47(+)	82.34±11.58(+)	**86.53±7.89**
±MU	30	74.24±13.27(+)	75.92±11.29(+)	77.59±10.36(-)	**77.80±9.18**(-)	76.97±10.34(+)	75.51±11.86(+)	75.70±9.72(+)	74.03±10.72(+)	75.10±12.76(+)	77.13±9.34
	40	75.26±10.18(+)	75.70±11.57(+)	76.32±9.65(+)	75.70±10.77(+)	76.32±8.45(+)	74.42±10.46(+)	75.77±6.67(+)	72.34±12.88(+)	75.68±8.70(+)	**76.53±10.44**
	50	75.06±10.49(+)	75.69±10.31(+)	**77.17±10.16**(-)	76.12±10.68(+)	76.95±8.65(+)	76.32±8.90(+)	76.52±9.90(+)	75.87±6.92(+)	75.47±10.00(+)	77.16±9.95
±FE	30	**89.92±9.22**(-)	87.48±7.90(-)	88.89±7.01(-)	80.38±3.79(-)	80.05±3.44(-)	86.44±9.06(-)	82.85±13.05(-)	84.79±13.10(-)	86.91±7.11(-)	79.87±3.86
	40	**88.61±6.58**(-)	**88.61±6.58**(-)	88.37±6.11(-)	83.11±4.05(+)	88.24±7.52(-)	88.37±7.22(-)	85.21±10.61(+)	85.12±10.70(+)	88.32±7.01(-)	87.69±8.98
	50	**88.56±7.08**(-)	88.28±5.93(-)	88.32±6.23(-)	83.20±4.05(+)	87.85±7.90(-)	88.32±585(-)	85.73±9.58(+)	85.87±9.52(+)	87.95±6.62(-)	87.63±10.32
±WI	30	92.22±8.31(=)	91.66±8.12(+)	**95.55±5.41**(-)	93.26±5.43(-)	81.43±8.18(+)	89.96±8.14(+)	82.02±8.13(+)	80.9±7.53(+)	91.11±8.67(+)	92.22±8.09
	40	96.66±2.72(-)	96.66±2.72(-)	**97.22±2.77**(-)	96.63±5.67(-)	94.93±3.90(+)	96.66±4.06(-)	92.15±6.64(+)	91.60±6.60(+)	96.66±3.68(-)	95.56±5.02
	50	**97.22±2.77**(-)	96.66±2.72(-)	96.66±2.72(-)	96.63±5.67(-)	96.66±4.43(-)	96.11±5.58(-)	93.82±5.30(+)	94.37±5.62(+)	96.66±2.72(-)	95.56±5.02
±LC	30	56.48±12.59(-)	**59.25±13.54**(-)	53.70±13.24(+)	47.22±11.36(+)	45.37±12.89(+)	53.70±13.24(+)	50.00±12.85(+)	51.85±12.79(+)	53.70±12.78(+)	54.62±12.88
	40	60.00±13.17(-)	65.83±13.38(-)	**69.17±13.20**(-)	39.67±12.47(+)	53.33±13.16(+)	57.50±13.03(+)	32.50±12.94(+)	55.03±17.95(+)	62.50±13.18	59.65±13.33
	50	62.50±13.18(+)	62.50±13.18(+)	**69.17±13.20**(-)	42.50±12.59(+)	53.33±13.16(+)	56.67±13.45(+)	32.91±12.27(+)	51.67±18.55(+)	59.17±13.30(+)	65.83±13.38
±PR	30	91.00±9.43(+)	91.00±7.00(+)	91.00±9.43(+)	88.18±8.56(+)	84.36±12.95(+)	91.00±9.43(+)	90.09±8.94(+)	91.00±9.43(+)	91.00±9.43(+)	**91.00±7.00**
	40	91.00±9.43(+)	90.02±8.94(+)	92.09±7.48(=)	90.02±8.94(+)	87.18±8.87(+)	92.09±7.48(=)	**93.01±7.81**(-)	**93.01±7.81**(-)	91.00±9.43(+)	92.09±7.48
	50	91.00±9.43(+)	**92.09±7.48**(-)	**92.09±7.48**(-)	89.09±8.38(+)	88.02±8.74(+)	90.02±8.94(+)	91.00±9.43(+)	**92.09±7.48**(-)	91.05±7.00(=)	91.05±7.00
±LY	30	87.11±9.80(=)	88.12±11.44(-)	**89.21±11.11**(-)	83.00± 10.96(+)	85.71±10.97(+)	85.00±10.21(+)	84.20±11.10(+)	84.20±11.10(+)	85.71±8.97(+)	87.11±9.80
	40	87.04±10.96(+)	87.04±10.96(+)	**88.11±10.52**(-)	85.03±9.17(+)	82.01±11.63(+)	88.01±11.63(-)	88.11±13.42(-)	86.88±14.62(+)	87.11±10.52(+)	87.11±9.80
	50	87.04±10.96(+)	87.04±10.96(+)	**88.11±10.52**(-)	85.03±9.17(+)	82.01±11.63(+)	86.02±10.16(+)	85.10±10.21(+)	85.10±10.21(+)	86.02±10.16(+)	87.11±9.80
±AR	30	69.44±3.97(+)	75.65±5.53(+)	77.79±5.75(+)	71.36±4.85(+)	68.73±4.28(+)	78.03±6.67(+)	70.65±6.34(+)	73.98±4.67(+)	76.60±4.89(+)	**78.50±5.38**
	40	68.97±4.84(+)	65.63±4.61(+)	71.59±6.34(+)	68.03±5.27(+)	65.87±4.47(+)	71.12±6.06(+)	66.35±3.86(+)	67.07±4.91(+)	70.16±5.84(+)	**73.98±4.67**
	50	68.97±4.84(+)	66.35±5.23(+)	71.35±5.83(+)	68.26±5.47(+)	65.40±4.36(+)	71.12±5.15(+)	65.87±6.71(+)	67.06±6.07(+)	70.88±5.61(+)	**71.54±4.88**
±CO	30	81.90±15.76(+)	83.57±14.92(+)	85.23±15.71(+)	83.8±14.93(+)	82.14±15.32(+)	80.23±14.60(+)	81.90±15.76(+)	81.90±15.76(+)	83.57±14.92(+)	**86.90±16.29**
	40	80.23±12.55(+)	80.23±12.55(+)	**81.90±13.89**(=)	78.80±13.23(+)	**81.90±13.89**(=)	**81.90±13.89**(=)	**81.90±13.89**(=)	**81.90±13.89**(=)	**81.90±13.89**(=)	**81.90±13.89**
	50	81.90±13.89(=)	81.90±13.89(=)	81.90±13.89(=)	77.14±11.37(+)	80.23±12.55(+)	81.90±13.89(=)	**83.55±14.92**(-)	81.90±13.8(=)	81.90±13.89(=)	81.90±13.89
±LE	30	**98.57±4.28**(=)	95.89±6.29(+)	**98.57±4.28**(=)	90.11±11.10(+)	97.14±5.71(+)	95.89±8.96(+)	94.64±10.74(+)	94.64±6.58(+)	**98.57±4.28**(=)	**98.57±4.28**
	40	95.71±9.14(+)	97.14±8.57(+)	94.46±9.32(+)	93.31±9.34(+)	90.53±13.87(+)	**97.14±5.71**(=)	94.64±6.58(+)	91.78±11.27(+)	**97.14±5.71**(=)	**97.14±5.71**
	50	95.71±9.14(+)	97.14±8.57(+)	97.14±8.57(+)	93.21±6.81(+)	93.21±9.34(+)	94.46±9.32(+)	95.89±6.29(+)	93.24±9.34(+)	**97.14±5.71**(=)	**97.14±5.71**

Table 8 continued

Dataset	Top (%) features	Chi-square	ANOVA	MI	MRMR	CMIM	MultiSURF	SURF*	MultiSURF*	ReliefF	μ-Relief
HS	30	78.31±9.75(+)	80.15±5.73(=)	**83.17±6.84**(-)	80.18±5.92(=)	77.93±4.67(+)	78.67±5.91(+)	80.15±4.05(+)	77.96±6.56(+)	77.16±6.70(+)	80.15±5.73
	40	78.23±7.24(+)	80.12±5.66(+)	80.10±7.53(+)	79.78±3.70(+)	77.90±2.55(+)	79.01±5.72(+)	77.53±5.02(+)	78.26±4.08(+)	79.01±7.11(+)	**80.18±4.57**
	50	77.84±8.01(+)	79.72±7.72(+)	**81.61±6.04**(-)	79.41±3.35(+)	77.90±2.55(+)	79.38±5.75(+)	77.89±4.62(+)	79.00±4.63(+)	77.85±9.05(+)	79.78±3.31
GL	30	87.91±9.35(+)	87.91±9.35(+)	87.91±9.35(+)	86.80±9.82(+)	85.41±12.08(+)	87.91±9.35(+)	84.30±11.16(+)	84.30±11.16(+)	**89.16±10.03**(=)	**89.16±10.03**
	40	**89.16±10.03**(=)	87.91±9.35(+)	87.91±9.35(+)	85.55±11.55(+)	86.67±11.57(+)	87.91±9.35(+)	84.30±11.16(+)	84.30±11.16(+)	**89.16±10.03**(=)	**89.16±10.03**
	50	**89.16±10.03**(=)	87.91±9.35(+)	87.91±9.35(+)	86.67±10.13(+)	86.67±11.57(+)	87.91±9.35(+)	85.41±12.08(+)	84.30±11.16(+)	87.91±9.35(+)	**89.16±10.03**
AM	30	97.14±5.71(+)	97.14±5.71(+)	**97.32±5.37**(=)	94.28±9.47(+)	93.03±13.08(+)	**97.32±5.37**(=)	88.92±10.61(+)	89.10±10.11(+)	**97.32±5.37**(=)	**97.32±5.37**
	40	**98.73±2.85**(=)	**98.73±2.85**(=)	**98.73±2.85**(=)	95.89±6.29(+)	95.89±6.29(+)	**98.73±2.85**(=)	90.17±12.88(+)	86.25±12.42(+)	**98.73±2.85**(=)	**98.73±2.85**
	50	**98.73±2.85**(-)	**98.73±2.85**(-)	97.32±5.37(+)	94.45±9.32(+)	95.89±6.29(+)	**98.73±2.85**(-)	88.75±13.97(+)	87.67±13.08(+)	**98.73±2.85**(-)	97.32±5.37

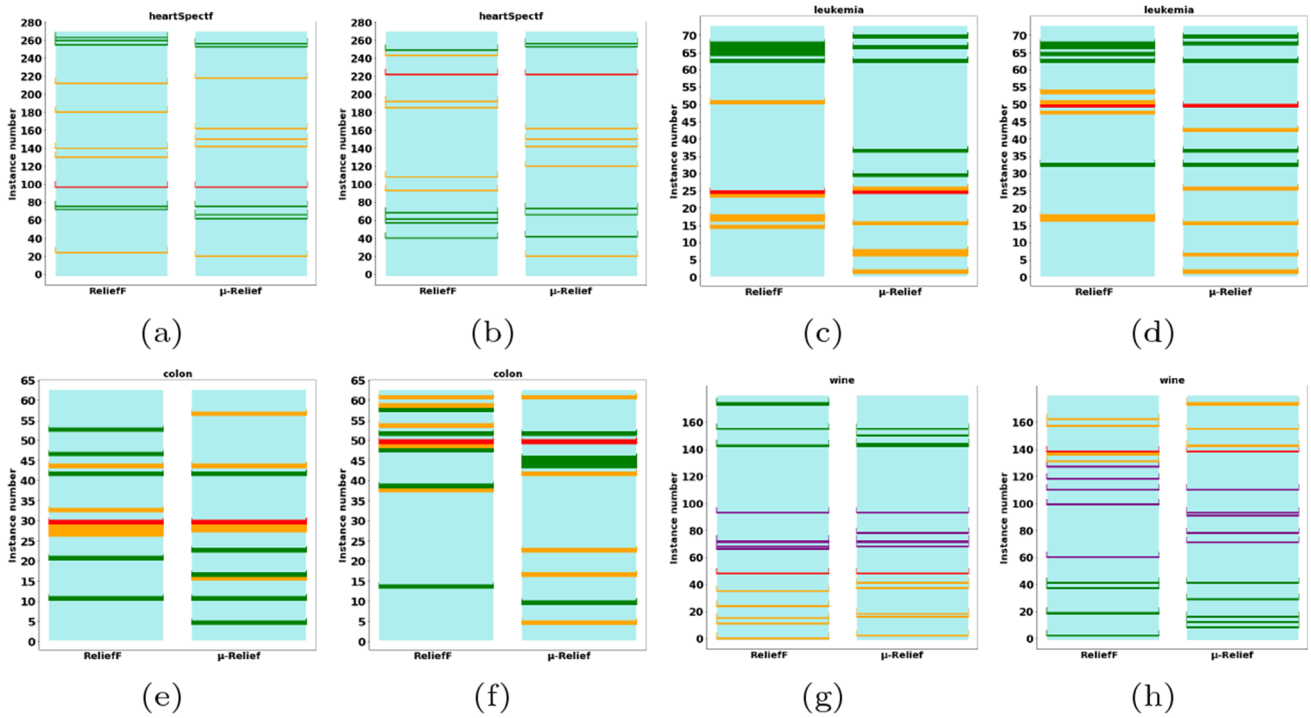

Fig. 4 Different instances selected for weight calculation by ReliefF and μ-Relief on real-world datasets

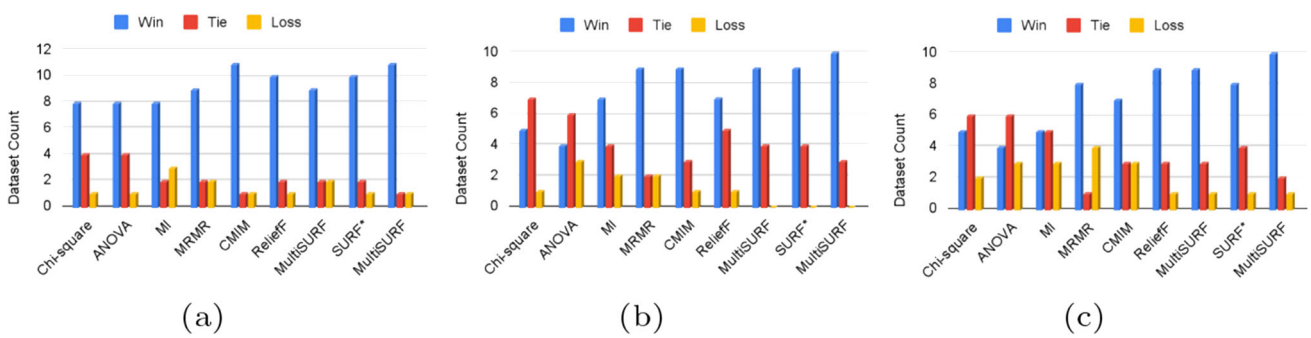

Fig. 5 μ-Relief performance as "better/equally-well/worse" compared to other methods with SVM on 30%,40% and 50% respectively

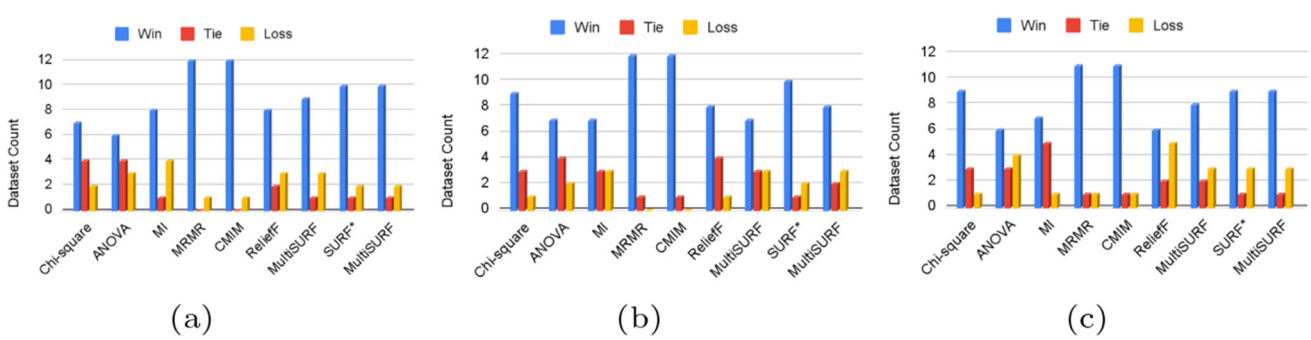

Fig. 6 μ-Relief performance as "better/equally-well/worse" compared to other methods with LR on 30%,40% and 50% respectively

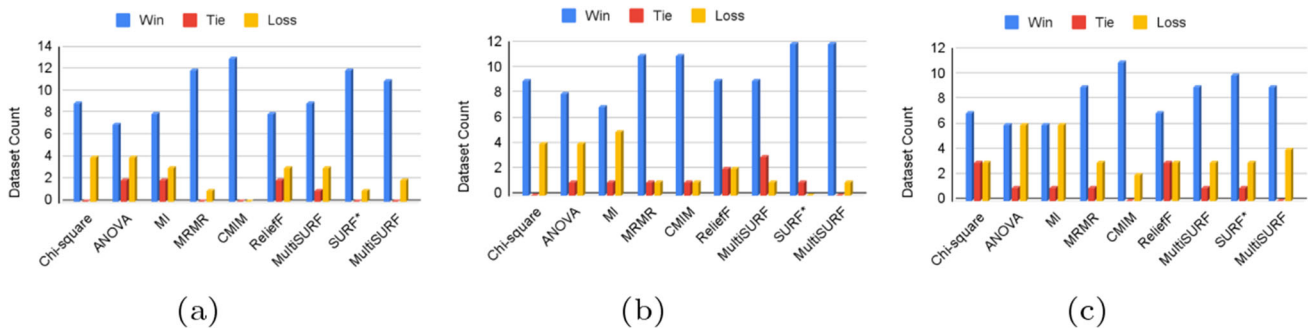

Fig. 7 μ-Relief performance as "better/equally-well/worse" compared to other methods with XGBRF on 30%,40% and 50% respectively

feature sets and 5 datasets for the 40% feature set. The best accuracy values using all feature sets of the *Parkinsons, musk, fetal* and *ALLAML* datasets were observed using Chi-square with XGBRF. μ-Relief performs better on 6 more data subsets as compared to the second-best method (Chi-square) and outperforms it by nearly 15%. As shown in Table 8, on the 30%, 40%, and 50% feature sets from 6, 8 and 4 datasets, respectively, the combination of μ-Relief with RF performed the best and outperformed the second-best classifier by 8%. μ-Relief was best on 18 out of 39 data subsets while MI was on 15 subsets. Seven methods have resulted in the same mean accuracy with RF for the *colon* dataset. The results in Table 5-8 depict that consideration of farthest neighbors makes μ-Relief perform better than other feature selection methods. Compared to ReliefF, the proposed algorithm performed better by around 49%, 39%, 31%, and 23% using SVM, LR, XGBRF and RF, respectively. Regardless of the number of features taken into account (30%, 40% or 50%), μ-Relief generally operates consistently.

The results of the two-tailed test on 13 datasets, Fig. 5(a), show that μ-Relief performed better than CMIM and MultiSURF* on 11 datasets for the top 30% features using SVM. With the same feature sets, μ-Relief yielded better mean accuracy than CMIM on 12 datasets with LR, XGBRF and RF classifiers, as shown in Figs. 6(a), 7(a), and 8(a) respectively. With the 50% feature set, μ-Relief, ANOVA and Chi-square methods tied for 6 datasets using the SVM algorithm, Fig. 5(c). Compared to MultiSURF*, μ-Relief performed better on 10 datasets for 40% and 50% feature sets using SVM, as reported in Fig. 5(b) and (c), respectively.

Using LR, the proposed method performed better than other methods for at least 6 datasets, as shown in Fig. 6. Remarkably better results were achieved with μ-Relief for the 40% feature set (Fig. 7(b) and 8(b)) with XGBRF and RF. Using the 50% feature set with the XGBRF classifier, μ-Relief, MI and ANOVA showed comparable performance (Fig. 7(c)). Compared to other methods, except for MI, superior results were obtained using μ-Relief with the 50% feature set (Fig. 8(c)). Using SVM and compared with ReliefF, μ-Relief scored 10, 7 and 9 wins on the 30%, 40%, and 50% feature sets, respectively (Fig. 5).

4.2.2 Technique-wise performance evaluation with number of features

Due to the limitation of visually indiscernible differences between the mean accuracy values with different feature selection methods using SVM, the number of most relevant features has been varied from 1 to a maximum value of 50. On the *leukemia*, *lymphoma, arrhythmia, prostrate*, *Parkinsons*, *ALLAML* and *colon* datasets, μ-Relief was more accurate and stable as shown in Fig. 9. On the fetal dataset, it achieved and maintained the highest mean accuracy after 19 features. On the *HeartSpectf* dataset, all the methods selected the same

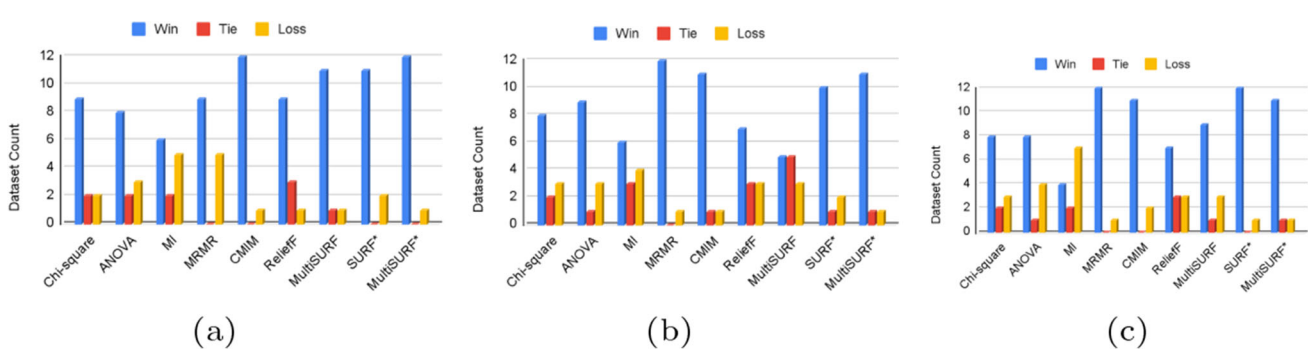

Fig. 8 μ-Relief performance as "better/equally-well/worse" compared to other methods with RF on 30%,40% and 50% respectively

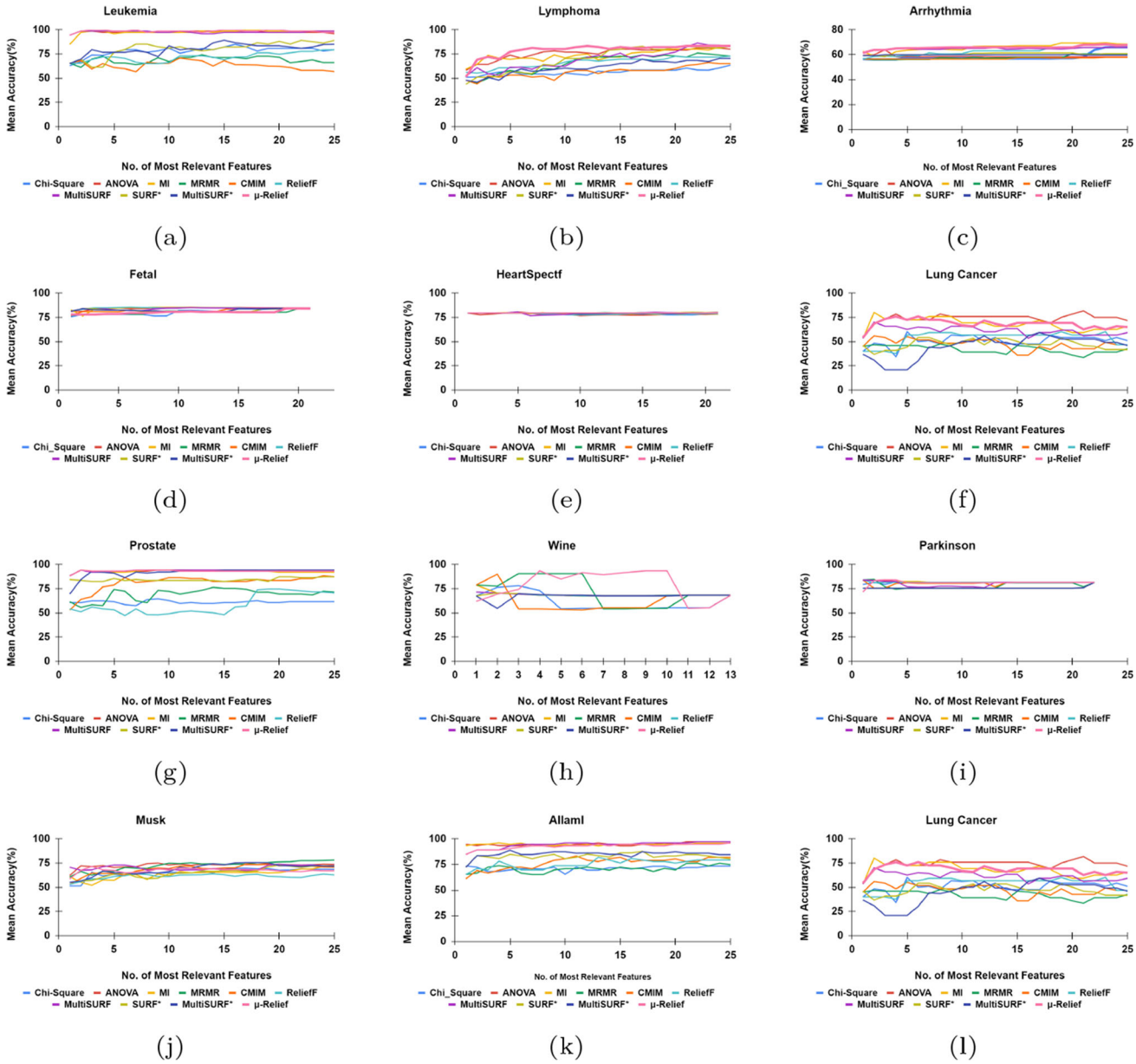

Fig. 9 Mean classification accuracy achieved on the real-world datasets with SVM classifier on 30% features set

features, and resulted in similar accuracies when the number of features was increased. The proposed method performed better than the 7 methods till the top 8 features with the *lung cancer* dataset. However, the average accuracy of all the methods fluctuated inconsistently when more features were considered. While MI and MRMR performed better with fewer features using the *wine* dataset, they declined with increasing dimensionality. On the contrary, μ-Relief provided more sustained accuracy peaks when it considered more features. From the majority of the datasets, μ-Relief selected the most relevant features when relatively fewer features were considered for classification.

Furthermore, other performance parameters obtained were described in Sect. 4.2.1. The F1-score (weighted) is a better performance measure when dealing with imbalanced datasets; it conveys the balance between precision and recall. Figure 10 shows the distribution of the F1-score of four classifiers using the top 30% features for all feature selection methods for all datasets. It is evident from Fig. 10(a) and (c) that the maximum, minimum, and median F1-scores of μ-Relief with SVM and XGBRF, are greater than or equal to the scores of other feature selection methods. The box plot in Fig. 10(b) indicates that the median score of the proposed method is as good as those of other state-of-the-art methods

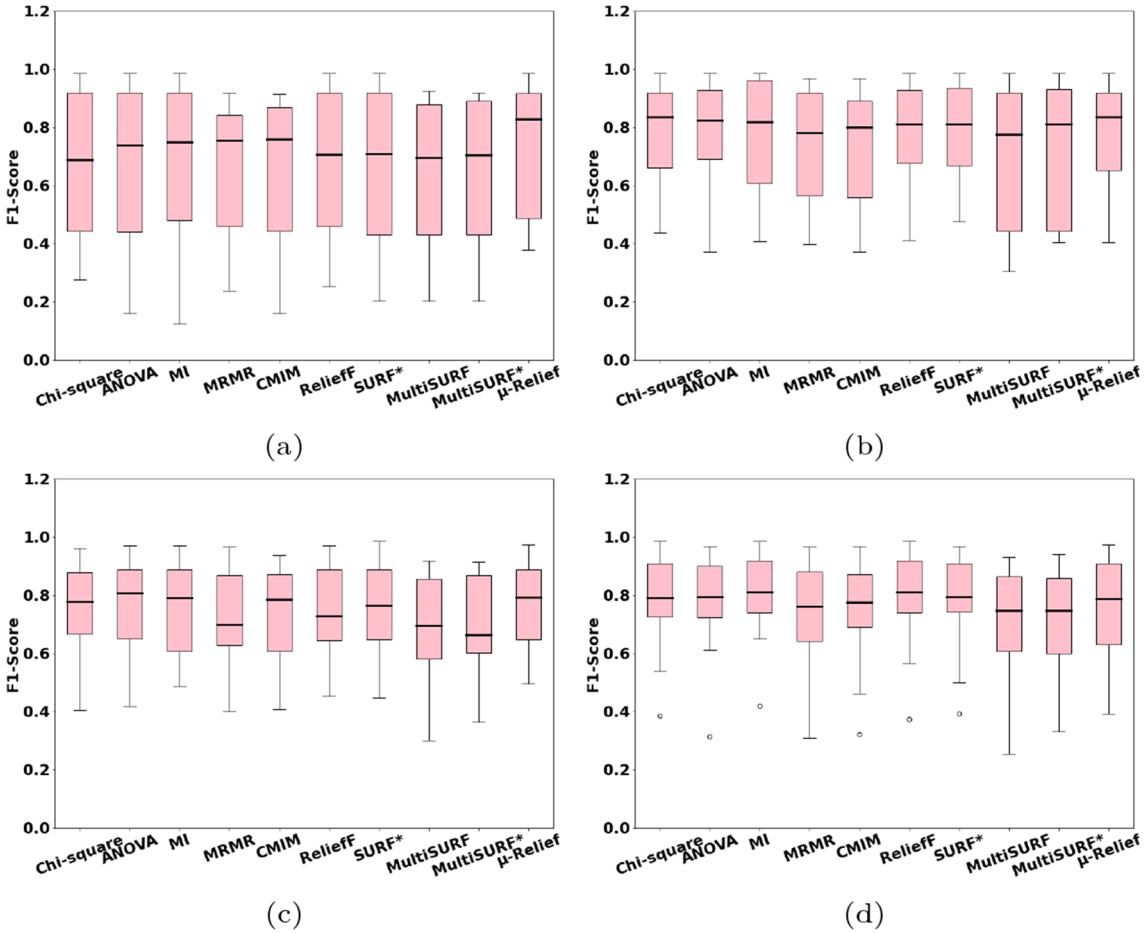

Fig. 10 Box-plot results for mean F1-score (weighted) of all the 10 feature selection method on the following algorithm **a** SVM, **b** LR, **c** XGBRF, **d** RF

using the LR classifier. Figure 10(d) shows that the maximum and median scores, computed using μ-Relief with RF, is the best when compared to other methods. It has also shown that 6 methods have their minimum scores as outliers. The distribution of ROC-AUC of the all four classifiers using the top 30% features from all datasets for all feature selection methods, as shown in Fig. 11, reveals that the classifiers trained on features selected using μ-Relief, were better than or comparable to those trained on features inferred by other methods.

5 Conclusion

This study proposed μ-Relief, a new filter feature selection method using the mean distance approach. Unlike ReliefF which uses the nearest neighbors approach, the μ-Relief follows a procedure of neighbor identification for calculation of feature scores based on mean distances. This strategy incorporates diversities of all features along with their information. This analysis focused on developing the new algorithm and comparing it with other state-of-art filter methods for feature selection. The study has performed on 9 filter methods using thirteen real-world high-dimensional datasets of different types from various domains. 30%, 40%, and 50% of the most relevant features were considered from each real-world dataset to assess the performance for each feature selection method individually. In addition, three synthetic datasets were generated and experiments were performed to evaluate the behavior of the proposed algorithm and its ability to assign better quality estimates to features. Through the experiments on synthetic datasets, the validity of μ-Relief as a standard feature selection method and its far-sightedness were confirmed. In addition, the differences between the selected neighbors using μ-Relief and ReliefF were also studied. The proposed algorithm includes diverse information of all features and effectively handles the non-uniform distribution of feature values. Classification mean accuracy, F1-score, and AUC-ROC were calculated for μ-Relief during the experiments on real-world datasets. These metrics were then compared with those of nine previously proposed algorithms, namely, ReliefF, SURF*, MultiSURF, MultiSURF*, Chi-square, ANOVA, MI, MRMR, and CMIM, using SVM,

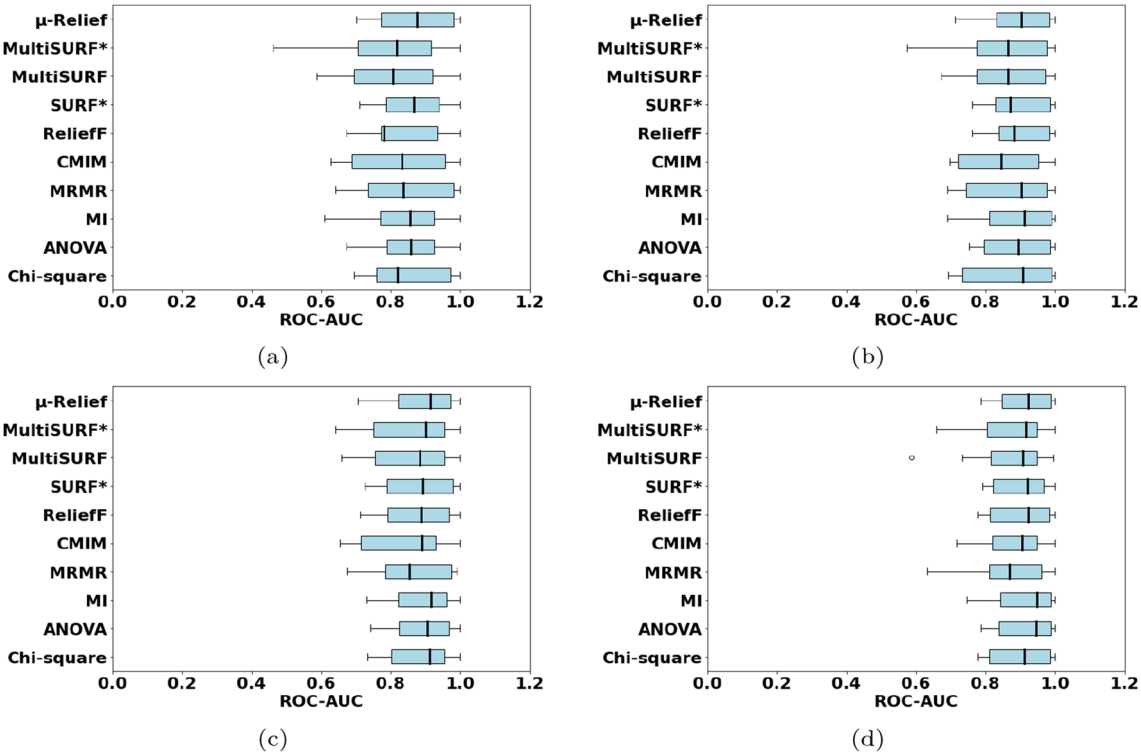

Fig. 11 Box-plots for ROC-AUC of all the 10 feature selection methods using **a** SVM, **b** LR, **c** XGBRF, **d** RF

LR, XGBRF, and RF as classifiers. μ-Relief has better mean accuracy as compared to ReliefF on 10 datasets and equivalent on 2 datasets with 30% feature set using SVM. With 40% and 50% feature set, μ-Relief has better mean accuracy on at least 7 and 9 datasets respectively with respect to ReliefF with the same classifier. Therefore, it can be concluded that μ-Relief is significantly advantageous than ReliefF for feature selection and has been selected for better features even on 30% features set. It is superior to MRMR and MultiSURF for at least 9 out of 13 real-world datasets with 30% feature set. Using the same feature set, the proposed algorithm performs better than SURF* and MultiSURF* for at least 10 real-world datasets and on 11 datasets when compared with CMIM. However, due to its fundamental design similar to an RBA, μ-Relief inherits the limitation of reduced performance on the feature set comprising features of mixed data types as visible for *fetal* dataset. We further plan to study the performance of this algorithm with noisy and unlabeled datasets with differing percentages of categorical and numerical features. The analysis and the novel approach presented in this paper shall serve as a reference to researchers to conduct feature filtering-based research.

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Declarations

Conflict of interest The authors have declared that they have no conflict of interest.

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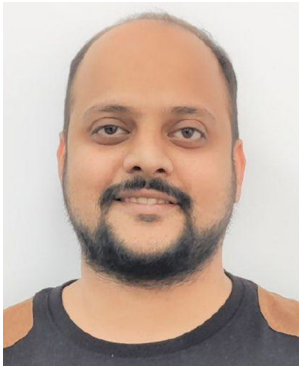

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